

Population Dynamics(508)

MD Salauddin

September 2026

Contents

1	Qualitative Theory	3
2	Periodic solutions	10
2.1	State and prove Bendixon's non-existence theorem.	12
2.2	Examine the existence of limit cycle and periodic solution of the system: If periodic solutions exist, find them and sketch the phase portrait.	13
2.3	Problem: Existence of a Limit Cycle for the system: 2021	16
3	Stability Theorem	17
3.1	The given system, Test whether the system stable or unstable	19
3.2	Discuss the stability and nature of the critical point.	20
3.3	Discuss the stability and nature of the critical point of the non linear system and sketch the phase portrait.	21
3.4	Discuss the stability and nature of the critical point of the non linear system and sketch the phase portrait.	22
3.5	Find critical point. Discuss the stability and nature of the critical point of the non linear system and sketch the phase portrait.	23
3.6	State the three theorems of Liapunov. 17	25
3.7	Examine the stability of the solution $\dot{x} = -x - y - x^3$ $\dot{y} = x - y - y^3$ by taking a suitable Liapunov-function. 2018	26
3.8	Obtain a Lyapunov function for the dynamical system	27
4	Simple species Population Model	29
4.1	Describe and solve the single species population model. Find the doubling/tripling time. Discuss the plausibility and stability conditions. (Malthusian)	30
4.2	Assume that, the population of Rajshahi city increases at a rate proportional to the number of inhabitants at any time t . If the population in double in 40 years, how many years it will be triple.	32
4.3	Describe logistic model/Verhulst model. Find its solution. Also find the ultimate/maximum population.	33
4.4	Find the equilibrium population of logistic model and discuss the stability.	35
4.5	Find the doubling time population for logistic model.	36
4.6	A population grows according to the model, $\frac{dN}{dt} = 1 - \exp \left[1 - r_0 \left(1 - \frac{N}{K} \right) \right],$ where r_0 and K are positive constants. Determine the equilibrium state and its stability.	37

4.7	The population N of Rajshahi satisfy the Logistic model,	
	$\frac{dN}{dt} = 0.03N - 3 \times 10^{-8}N^2,$	
	where t is measured in years. If the population of Rajshahi was 2,00,000 in 1980. What will be the population in the year 2000. What would be the possible maximum pop ⁿ of Rajshahi.	38
4.8	If the carrying capacity of the logistic model is K and $0 < N_0 < \frac{K}{2}$, then determine the time of doubling of population.2018	40
4.9	The world's population was estimated to be 5000 million in 1950 and 7500 million in 2000. Estimate the population of the world in the year 2050, given that the population increases according to the Malthusian model.2023	41
4.10	Suppose the population of Bangladesh grows according to the Malthusian model. The population of Bangladesh was estimated to be 170.00 million in 2010 and 180.00 million in 2020. Estimate the population of Bangladesh in the year 2030 and also in 2040.2021	42
4.11	If $\frac{dP}{dt} = rP - k$, $P(0) = P_0$, then solve this equation and make a realistic comment regarding the solution.2020	44
5	Discrete population Model	46
5.1	Describe the Harvesting Population Model and find its solutions. Discuss the stability of the Harvesting Model.	47
5.2	Determine the extinction time and tripling time population govern by the Model	50
5.3	Describe the two species linear competition model and identify it.(not CL)	52
5.4	Describe the symbiotic population model and explain it.(not CL)	52
5.5	The population $N(t)$ (in million) of Bangladesh satisfies the model $\frac{dN}{dt} = \alpha N - \beta N^2$ where t is measured in years. If $N(2000) = 127.66$, $N(2010) = 147.56$ and $N(2020) = 164.67$, find the population of Bangladesh in 2030.(2020)	53
6	Two species Population Model	56
6.1	Describe the two species linear predator prey model and explain it. . . .	57
6.2	Identify and solve the two species linear population model.	58
6.3	State and explain the Latka-Volterra predator prey model	59
6.4	Problem: Two-Species Competition Model2021	65
6.5	Identify and solve the two-species linear population model:	67
7	Epidemic/Infection disease Model	70
7.1	Describe the simple SIR epidemic model. Find the condition on which the infection will ultimately die out or spread throughout the population. . .	71
7.2	Discuss the deterministic susceptible infected(SI) model and solve. Show that the infection will spread throughout the population.	73
8	HIV	76
8.1	Describe the deterministic AIDS model. Show that the infection spread throughout the population. Find the doubling time. Also find the number of AIDS patients at any time t	76
8.2	For understanding	78

Chapter 1

Qualitative Theory

Dynamical System

A system of equations:

$$\begin{aligned}\frac{dx}{dt} &= P(x, y) \\ \frac{dy}{dt} &= Q(x, y)\end{aligned}$$

which represents a physical problem is called a **dynamical system**.

Example:

Consider the system of equations:

$$\begin{aligned}\frac{dx}{dt} &= x - y \\ \frac{dy}{dt} &= -x + y\end{aligned}$$

This is a dynamical system.

Autonomous and Non-Autonomous Dynamical Systems

An **autonomous dynamical system** is a system of differential equations in which the independent variable time t does not explicitly appear in the defining functions.

Formulation

In a two-dimensional system, it is represented as:

$$\begin{aligned}\dot{x} &= \frac{dx}{dt} = P(x, y) \\ \dot{y} &= \frac{dy}{dt} = Q(x, y)\end{aligned}$$

where the functions P and Q depend solely on the state variables x and y , and not directly on t .

Non-Autonomous Dynamical System

If the independent variable t appears explicitly in the governing equations, the system is called a **non-autonomous system**:

$$\begin{aligned}\dot{x} &= \frac{dx}{dt} = P(x, y, t) \\ \dot{y} &= \frac{dy}{dt} = Q(x, y, t)\end{aligned}$$

Examples

1. Autonomous System:

$$\begin{aligned}\dot{x} &= xy \\ \dot{y} &= x^2 + y^2\end{aligned}$$

(No explicit time parameter t on the right-hand side.)

2. Non-Autonomous System:

$$\begin{aligned}\dot{x} &= xt + y \\ \dot{y} &= x^2 + y^2 + t^2\end{aligned}$$

(Time parameter t explicitly appears on the right-hand side.)

Critical Point

Let

$$\dot{x} = P(x, y), \quad \dot{y} = Q(x, y) \tag{1}$$

be a dynamical system. A point (x_0, y_0) is called a **critical point** of the system (1) if:

$$(i) \quad P(x_0, y_0) = 0 \quad \text{and} \quad Q(x_0, y_0) = 0$$

i.e.,

$$\begin{aligned}\dot{x} &= 0 \\ \dot{y} &= 0\end{aligned}$$

Isolated Critical Points

Let

$$\dot{x} = P(x, y) \quad \text{and} \quad \dot{y} = Q(x, y) \tag{i}$$

be a dynamical system, and let (x_0, y_0) be a critical point of (i). If there exists a neighborhood of the critical point (x_0, y_0) such that no other critical point lies in that neighborhood, then (x_0, y_0) is called an **isolated critical point**.

There are four types of isolated critical points:

- (i) Node
- (ii) Saddle point
- (iii) Spiral (Focus)
- (iv) Center

Node:

Consider the autonomous dynamical system:

$$\dot{x} = P(x, y), \quad \dot{y} = Q(x, y)$$

with an isolated critical point at the origin $(0,0)$. The critical point $(0,0)$ is called a **node** if:

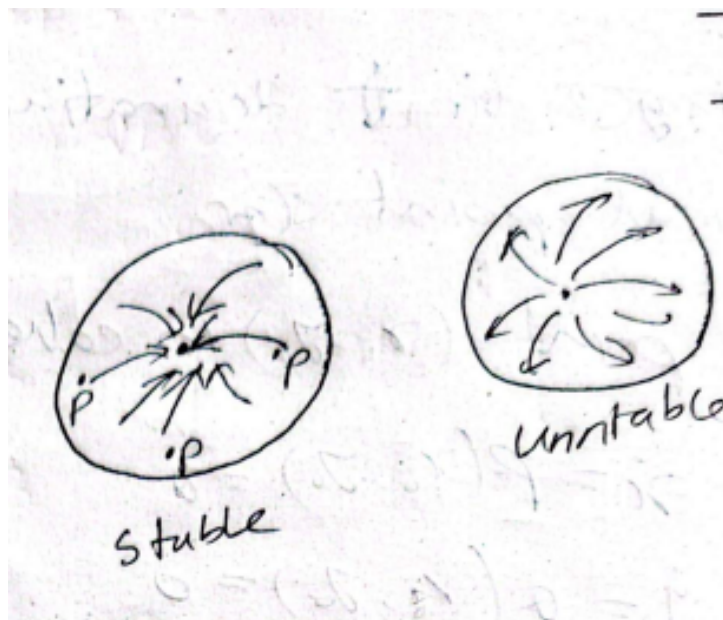
1. Every trajectory in a neighborhood of $(0,0)$ approaches $(0,0)$ either as $t \rightarrow +\infty$ (stable node) or as $t \rightarrow -\infty$ (unstable node).
2. Every trajectory enters or leaves $(0,0)$ along a **well-defined tangent line** (a definite direction).
3. Trajectories **do not spiral or rotate** infinitely around $(0,0)$ as they approach or leave it.

Alternative (Eigenvalue Criterion for Linear Systems)

If the condition is given in terms of linear system dynamics ($\dot{\mathbf{x}} = A\mathbf{x}$):

*The isolated critical point $(0,0)$ is a **node** if the eigenvalues λ_1, λ_2 of the Jacobian matrix evaluated at $(0,0)$ are **real, non-zero, distinct (or proper)**, and have the same sign:*

- **Stable Node:** $\lambda_1, \lambda_2 < 0$ (Real and both negative)
- **Unstable Node:** $\lambda_1, \lambda_2 > 0$ (Real and both positive)



Definition of a Saddle Point

Consider the autonomous dynamical system $\dot{x} = P(x, y)$, $\dot{y} = Q(x, y)$ with an isolated critical point at $(0, 0)$.

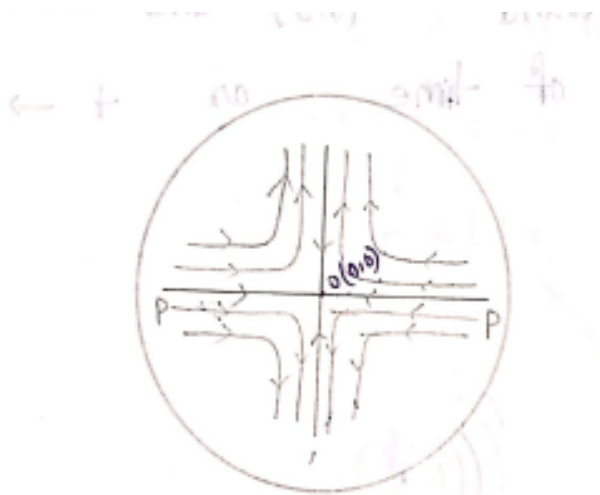
The critical point $(0, 0)$ is called a **saddle point** if:

1. There exist exactly **two trajectories** (separatrices) that approach and enter $(0, 0)$ as $t \rightarrow +\infty$ from opposite directions, and **two trajectories** that enter $(0, 0)$ as $t \rightarrow -\infty$ (move away as $t \rightarrow +\infty$) from opposite directions.
2. All other trajectories in the neighborhood of $(0, 0)$ pass near the critical point without entering it and subsequently move away as $t \rightarrow \pm\infty$ (forming hyperbolic-like curves).

Eigenvalue Condition (Linear Analysis)

For the linearized system $\dot{\mathbf{x}} = A\mathbf{x}$, $(0, 0)$ is a **saddle point** if and only if the eigenvalues λ_1, λ_2 of the Jacobian matrix evaluated at $(0, 0)$ are **real, non-zero, and have opposite signs**:

$$\lambda_1 < 0 < \lambda_2 \quad (\text{or } \det(A) < 0)$$



Definition of a Spiral (or Focus)

Consider the autonomous dynamical system $\dot{x} = P(x, y)$, $\dot{y} = Q(x, y)$ with an isolated critical point at $(0, 0)$.

The critical point $(0, 0)$ is called a **spiral** (or **focus**) if:

1. Trajectories starting at any point P in a neighborhood of $(0, 0)$ approach the origin as $t \rightarrow +\infty$ (stable spiral) or move away from the origin as $t \rightarrow +\infty$ (unstable spiral).

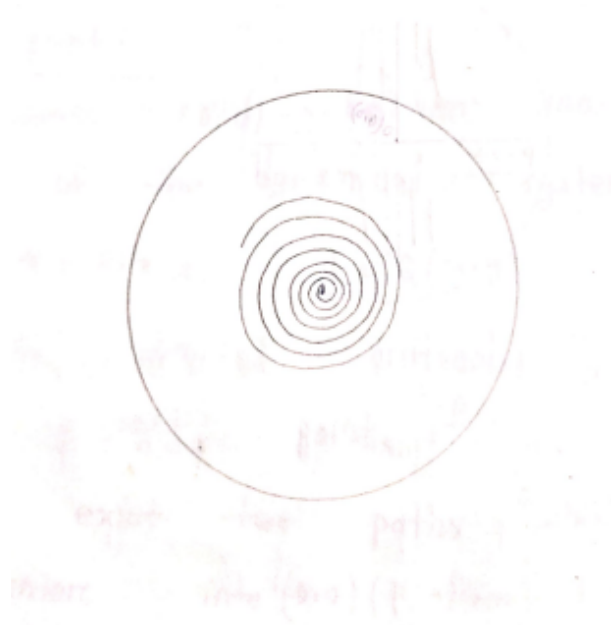
2. Trajectories wind around $(0, 0)$ **infinitely many times** in a spiral-like manner as they approach or move away from the critical point.
3. Trajectories **do not enter or leave along a single well-defined tangent line**, as the direction angle $\theta(t) \rightarrow \pm\infty$.

Eigenvalue Condition (Linear Analysis)

For the linearized system $\dot{\mathbf{x}} = A\mathbf{x}$, $(0, 0)$ is a **spiral point** if and only if the eigenvalues λ_1, λ_2 of the Jacobian matrix A evaluated at $(0, 0)$ are **complex conjugates with a non-zero real part**:

$$\lambda_{1,2} = \alpha \pm i\beta, \quad \text{where } \alpha \neq 0 \text{ and } \beta \neq 0$$

- **Stable Spiral (Sink):** $\alpha = \text{Re}(\lambda) < 0$ (trajectories spiral inward toward $(0, 0)$).
- **Unstable Spiral (Source):** $\alpha = \text{Re}(\lambda) > 0$ (trajectories spiral outward away from $(0, 0)$).



Definition of a Center

Consider the autonomous dynamical system $\dot{x} = P(x, y)$, $\dot{y} = Q(x, y)$ with an isolated critical point at $(0, 0)$.

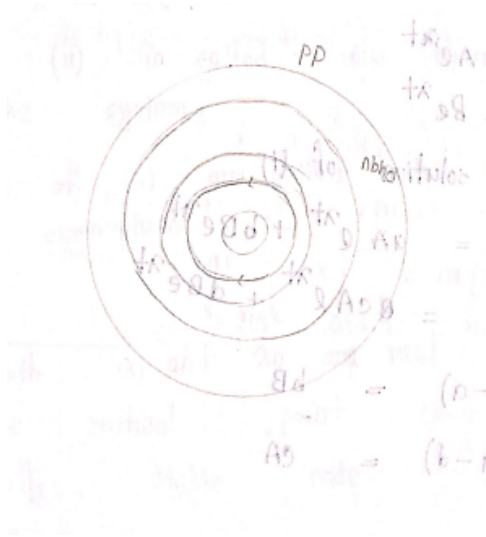
The critical point $(0, 0)$ is called a **center** (or **centre**) if:

1. Every neighborhood of $(0, 0)$ contains a family of **closed concentric trajectories** (ellipses or circles) enclosing $(0, 0)$ as an interior point.
2. Trajectories around $(0, 0)$ are periodic and closed, meaning they **neither approach nor move away** from $(0, 0)$ as $t \rightarrow \pm\infty$.

Eigenvalue Condition (Linear Analysis)

For the linearized system $\dot{\mathbf{x}} = A\mathbf{x}$, $(0, 0)$ is a **center** if and only if the eigenvalues λ_1, λ_2 of the Jacobian matrix A evaluated at $(0, 0)$ are **purely imaginary**:

$$\lambda_{1,2} = \pm i\omega \quad (\text{where } \omega \in \mathbb{R}, \omega \neq 0)$$



Chapter 2

Periodic solutions

Info

- Green's Theorem

Let C be a positively oriented (counterclockwise), piecewise-smooth, simple closed curve in the xy -plane, and let D be the region bounded by C . If $P(x, y)$ and $Q(x, y)$ are functions that have continuous first-order partial derivatives ($\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$) on an open region containing D , then: Line Integral / Double Integral Form (Standard Form)

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

where $dA = dx dy$ (or $dy dx$).

- **Angular equation:**

$$r^2 \dot{\theta} = x\dot{y} - y\dot{x}$$

Closed Path

If the initial and terminal points of a path are identical, then the path is called a **closed path**.

Examples of Closed Paths:

- A circle or ellipse returning to its starting point.

Non-Closed Path

If the initial and terminal points of a path are not identical, then the path is called a **non-closed path** (or open path).

Example of a Non-Closed Path:

- An open curve or line segment where the start and end points do not coincide.

Periodic Solution

Consider the dynamical system:

$$\begin{aligned}\dot{x} &= P(x, y) \\ \dot{y} &= Q(x, y)\end{aligned}\tag{1}$$

Let $x = f(t)$ and $y = g(t)$ be a solution of system (1). Then the solution is called a **periodic solution** if there exists a constant $T > 0$ such that:

$$f(t + T) = f(t) \quad \text{and} \quad g(t + T) = g(t) \quad \text{for all } t$$

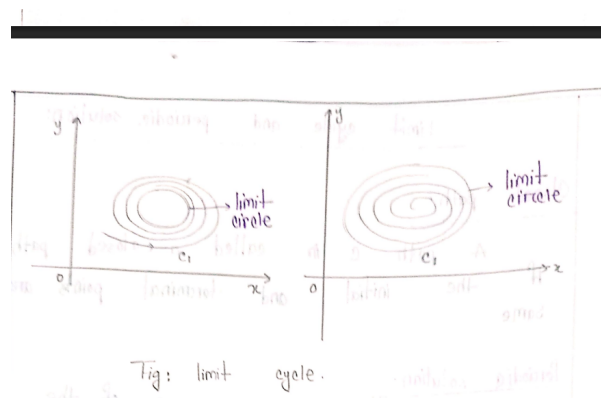
Here, T is called the **period** of the solution.

Limit Cycle

A closed path C of a system:

$$\begin{aligned}\dot{x} &= P(x, y) \\ \dot{y} &= Q(x, y)\end{aligned}\tag{1}$$

which is approached spirally from either inside or outside by a non-closed path C_1 of system (1) as $t \rightarrow +\infty$ or $t \rightarrow -\infty$ is called a **limit cycle** of the system (1).



2.1 State and prove Bendixon's non-existence theorem.

Statement

If $P(x, y)$ and $Q(x, y)$ of the system:

$$\begin{aligned}\dot{x} &= P(x, y) \\ \dot{y} &= Q(x, y)\end{aligned}\tag{1}$$

have continuous first-order partial derivatives in a simply-connected domain D of the xy -plane, and if $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ has the same sign throughout D , then system (1) has no closed path (periodic solution) in D .

Proof

Let C be a closed path in D and R be the region bounded by C .

Then by Green's theorem in the plane, we have:

$$\oint_C [P(x, y) dy - Q(x, y) dx] = \iint_R \left[\frac{\partial P(x, y)}{\partial x} + \frac{\partial Q(x, y)}{\partial y} \right] dx dy \tag{2}$$

where the line integral is taken in the positive (counterclockwise) sense.

Now, assume that C is a closed path of system (1).

Let $x = f(t)$, $y = g(t)$ be a periodic solution of system (1) representing the closed path C with period T .

Along the trajectory C :

$$\begin{aligned}\dot{x} &= \frac{dx}{dt} = \frac{df(t)}{dt} = P(x, y) = P(f(t), g(t)) \\ \dot{y} &= \frac{dy}{dt} = \frac{dg(t)}{dt} = Q(x, y) = Q(f(t), g(t))\end{aligned}\tag{3}$$

Substituting (3) into the Left-Hand Side (L.H.S.) of equation (2):

$$\begin{aligned}\text{L.H.S. of (2)} &= \oint_C [P(x, y) dy - Q(x, y) dx] \\ &= \int_0^T \left[P(f(t), g(t)) \frac{dg}{dt} - Q(f(t), g(t)) \frac{df}{dt} \right] dt \\ &= \int_0^T [P(f(t), g(t)) \cdot Q(f(t), g(t)) - Q(f(t), g(t)) \cdot P(f(t), g(t))] dt \quad [\text{Using (3)}] \\ &= 0\end{aligned}$$

Substituting this result back into equation (2) yields:

$$\iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy = 0 \tag{4}$$

But this double integral can be zero only if $\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right)$ changes sign in R . This is a contradiction. Thus, C is not a path of system (i). Hence, the system (i) has no closed path or periodic solution in D .

2.2 Examine the existence of limit cycle and periodic solution of the system: If periodic solutions exist, find them and sketch the phase portrait.

$$\begin{aligned}\dot{x} &= -y + x(1 - x^2 - y^2) \\ \dot{y} &= x + y(1 - x^2 - y^2)\end{aligned}$$

Solution:

Consider the non-linear autonomous dynamical system:

$$\begin{aligned}\dot{x} &= -y + x(1 - x^2 - y^2) = P(x, y) \\ \dot{y} &= x + y(1 - x^2 - y^2) = Q(x, y)\end{aligned}$$

now, $P(x, y) = -y + x(1 - x^2 - y^2)$ and $Q(x, y) = x + y(1 - x^2 - y^2)$.
Computing partial derivatives:

$$\begin{aligned}\frac{\partial P}{\partial x} &= (1 - x^2 - y^2) + x(-2x) = 1 - 3x^2 - y^2 \\ \frac{\partial Q}{\partial y} &= (1 - x^2 - y^2) + y(-2y) = 1 - x^2 - 3y^2\end{aligned}$$

Calculating the divergence of the vector field:

$$\nabla \cdot \mathbf{f} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 2 - 4x^2 - 4y^2 = 2[1 - 2(x^2 + y^2)]$$

Since $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ changes sign, Bendixson's negative criterion does not rule out the existence of closed trajectories. Using the transformation $x = r \cos \theta$ and $y = r \sin \theta$, where $r^2 = x^2 + y^2$: Differentiating $r^2 = x^2 + y^2$ with respect to t :

$$2r\dot{r} = 2x\dot{x} + 2y\dot{y} \implies r\dot{r} = x\dot{x} + y\dot{y}$$

Substituting $\dot{x}$ and $\dot{y}$:

$$\begin{aligned}x\dot{x} + y\dot{y} &= x[-y + x(1 - r^2)] + y[x + y(1 - r^2)] \\ &= -xy + x^2(1 - r^2) + xy + y^2(1 - r^2) \\ &= (x^2 + y^2)(1 - r^2) = r^2(1 - r^2)\end{aligned}$$

Equating $r\dot{r} = r^2(1 - r^2)$ (for $r \neq 0$):

$$\dot{r} = r(1 - r^2)$$

Angular Equation ($\dot{\theta}$)

Using $r^2\dot{\theta} = x\dot{y} - y\dot{x}$:

$$\begin{aligned}x\dot{y} - y\dot{x} &= x[x + y(1 - r^2)] - y[-y + x(1 - r^2)] \\ &= x^2 + xy(1 - r^2) + y^2 - xy(1 - r^2) \\ &= x^2 + y^2 = r^2\end{aligned}$$

Equating $r^2\dot{\theta} = r^2$:

$$\dot{\theta} = 1$$

$$\frac{d\theta}{dt} = 1 \implies \theta(t) = t + \theta_0$$

Setting phase angle $\theta_0 = 0$ for simplicity gives $\theta(t) = t$.

Separating variables in $\frac{dr}{dt} = r(1 - r^2)$:

$$\frac{dr}{r(1 - r^2)} = dt$$

Applying partial fraction decomposition:

$$\int \left(\frac{1}{r} + \frac{r}{1 - r^2} \right) dr = \int dt$$

$$\ln r - \frac{1}{2} \ln |1 - r^2| = t + k$$

$$\ln \left(\frac{r}{\sqrt{|1 - r^2|}} \right) = t + k \implies \frac{r^2}{|1 - r^2|} = C_1 e^{2t} \quad (C_1 = e^{2k})$$

Solving for r^2 :

$$r^2 = \frac{C_0 e^{2t}}{1 + C_0 e^{2t}} = \frac{1}{1 + C e^{-2t}} \quad \left(C = \frac{1}{C_0} \right)$$

With initial condition $r(0) = r_0$, we obtain $C = \frac{1 - r_0^2}{r_0^2}$:

$$r(t) = \frac{1}{\sqrt{1 + \left(\frac{1 - r_0^2}{r_0^2} \right) e^{-2t}}}$$

$$\therefore x(t) = r(t) \cos(t) = \frac{\cos t}{\sqrt{1 + \left(\frac{1 - r_0^2}{r_0^2} \right) e^{-2t}}}$$

$$\text{and } y(t) = r(t) \sin(t) = \frac{\sin t}{\sqrt{1 + \left(\frac{1 - r_0^2}{r_0^2} \right) e^{-2t}}}$$

1. Periodic Orbit / Limit Cycle ($r_0 = 1$):

If $r_0 = 1$, then $r(t) = 1$ for all $t \geq 0$. The solution simplifies to:

$$x(t) = \cos t, \quad y(t) = \sin t$$

This represents an isolated closed trajectory (a circle of radius $r = 1$) in the phase plane.

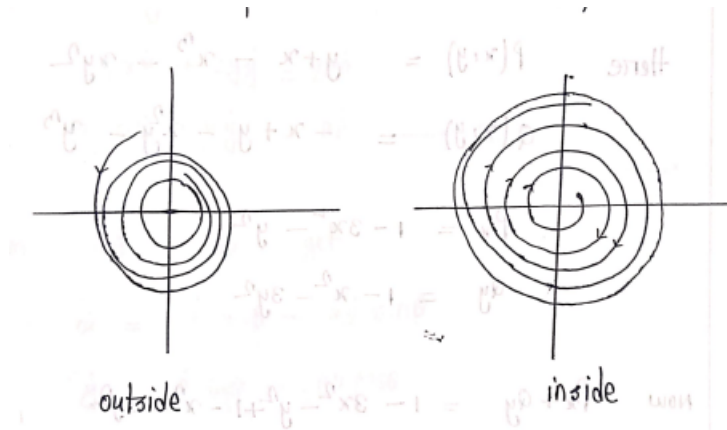
2. Outer Trajectories ($r_0 > 1$):

As $t \rightarrow +\infty$, $e^{-2t} \rightarrow 0$, which yields $r(t) \rightarrow 1^-$. Trajectories starting outside the unit circle spiral inward toward $r = 1$.

3. Inner Trajectories ($0 < r_0 < 1$):

As $t \rightarrow +\infty$, $e^{-2t} \rightarrow 0$, which yields $r(t) \rightarrow 1^+$. Trajectories starting inside the unit circle spiral outward toward $r = 1$.

Conclusion: Because all nearby trajectories approach $r = 1$ as $t \rightarrow +\infty$ from both interior and exterior regions, $r = 1$ is a **stable limit cycle**.



2.3 Problem: Existence of a Limit Cycle for the system: 2021

$$\begin{aligned}\dot{x} &= 4x - 4y - x(x^2 + y^2) \\ \dot{y} &= 4x + 4y - y(x^2 + y^2)\end{aligned}$$

and verify that $x = 2 \cos(4t)$, $y = 2 \sin(4t)$ is a periodic solution of this system.

Solution:

Given system of differential equations:

$$\dot{x} = 4x - 4y - x(x^2 + y^2) \quad (1a)$$

$$\dot{y} = 4x + 4y - y(x^2 + y^2) \quad (1b)$$

Let $P(x, y) = 4x - 4y - x(x^2 + y^2)$ and $Q(x, y) = 4x + 4y - y(x^2 + y^2)$.

Computing partial derivatives:

$$\begin{aligned}\frac{\partial P}{\partial x} &= 4 - 3x^2 - y^2 \\ \frac{\partial Q}{\partial y} &= 4 - x^2 - 3y^2\end{aligned}$$

Taking the sum (divergence of the vector field):

$$\begin{aligned}\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} &= (4 - 3x^2 - y^2) + (4 - x^2 - 3y^2) \\ &= 8 - 4(x^2 + y^2) \quad (\text{divergence changes sign at } r = \sqrt{2})\end{aligned}$$

Since $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ changes sign in the plane, Bendixson's criterion does not rule out periodic orbits. Thus, the system admits a limit cycle.

Verification of Periodic Solution:

Given proposed solution:

$$x(t) = 2 \cos(4t) \quad (a)$$

$$y(t) = 2 \sin(4t)$$

Differentiating with respect to t :

$$\dot{x} = -8 \sin(4t) \quad (b)$$

$$\dot{y} = 8 \cos(4t)$$

Substituting (a) into the right-hand side of equation (1a):

$$\begin{aligned}\text{RHS} &= 4(2 \cos 4t) - 4(2 \sin 4t) - 2 \cos 4t (4 \cos^2 4t + 4 \sin^2 4t) \quad (x^2 + y^2 = 4) \\ &= 8 \cos 4t - 8 \sin 4t - 2 \cos 4t(4) \\ &= -8 \sin 4t = \dot{x} \text{RHS} \\ &= 4(2 \cos 4t) + 4(2 \sin 4t) - 2 \sin 4t (4 \cos^2 4t + 4 \sin^2 4t) \\ &= 8 \cos 4t + 8 \sin 4t - 2 \sin 4t(4) \\ &= 8 \cos 4t = \dot{y}\end{aligned}$$

Hence, $x(t) = 2 \cos(4t)$, $y(t) = 2 \sin(4t)$ is a periodic solution of the system.

There are two math problems in page around 110 on periodic solution, you should look on it.

Chapter 3

Stability Theorem

ABC

- positive definite
- Positive semi-definite
- Negative definite
- Negative semi-definite
- Liapunov function
- Trivial solution
- system of a non linear equation

Poincaré–Liapunov Theorem on Linearization

Statement

Let the non-linear autonomous system be given by:

$$\begin{aligned}\dot{x} &= ax + by + P(x, y) \\ \dot{y} &= cx + dy + Q(x, y)\end{aligned}\tag{1}$$

If $P(x, y)$ and $Q(x, y)$ are continuous in a neighborhood of the origin $(0, 0)$ and satisfy the conditions:

$$\begin{aligned}\lim_{(x,y) \rightarrow (0,0)} \frac{P(x, y)}{\sqrt{x^2 + y^2}} &= 0 \\ \lim_{(x,y) \rightarrow (0,0)} \frac{Q(x, y)}{\sqrt{x^2 + y^2}} &= 0\end{aligned}$$

then the characteristic nature and qualitative stability of the critical point at the origin $(0, 0)$ for the non-linear system (1) is the same as that of the corresponding linearized system:

- Squeeze Theorem:

Let $f(x, y)$, $g(x, y)$, and $h(x, y)$ be functions defined on a neighborhood of (x_0, y_0) , except possibly at (x_0, y_0) itself. If for all $(x, y) \neq (x_0, y_0)$ in that neighborhood:

$$g(x, y) \leq f(x, y) \leq h(x, y)$$

and if:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} g(x, y) = L \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} h(x, y) = L$$

then:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = L$$

For example:

$$\left| \frac{-x^2}{\sqrt{x^2 + y^2}} \right| = \frac{|x| \cdot |x|}{\sqrt{x^2 + y^2}} \leq \frac{|x| \cdot \sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} = |x| \quad \left[|x| \leq \sqrt{x^2 + y^2} \right]$$

Thus:

$$0 \leq \left| \frac{-x^2}{\sqrt{x^2 + y^2}} \right| \leq |x|$$

Taking the limit as $(x, y) \rightarrow (0, 0)$:

$$\lim_{(x,y) \rightarrow (0,0)} 0 = 0 \quad \text{and} \quad \lim_{(x,y) \rightarrow (0,0)} |x| = 0$$

By the Squeeze Theorem:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{-x^2}{\sqrt{x^2 + y^2}} = 0$$

3.1 The given system ,Test whether the system stable or unstable

$$\begin{aligned}\dot{x} &= ax + by \\ \dot{y} &= cx + dy\end{aligned}$$

Solution

Consider the two-dimensional linear homogeneous autonomous system:

$$\begin{aligned}\dot{x} &= ax + by \\ \dot{y} &= cx + dy\end{aligned}\tag{1}$$

Assume a trial solution of the form:

$$x(t) = Ae^{\lambda t}, \quad y(t) = Be^{\lambda t}$$

where A and B are non-zero constants.

Differentiating and substituting into system (1):

$$\begin{aligned}A\lambda e^{\lambda t} &= aAe^{\lambda t} + bBe^{\lambda t} \\ B\lambda e^{\lambda t} &= cAe^{\lambda t} + dBe^{\lambda t}\end{aligned}$$

Since $e^{\lambda t} \neq 0$ for all t :

$$\begin{aligned}A\lambda &= aA + bB \\ B\lambda &= cA + dB\end{aligned}$$

Rearranging into a homogeneous linear system:

$$\begin{aligned}(\lambda - a)A - bB &= 0 \\ -cA + (\lambda - d)B &= 0\end{aligned}$$

For non-trivial solutions $(A, B) \neq (0, 0)$, the determinant of the coefficient matrix must vanish:

$$\begin{vmatrix} \lambda - a & -b \\ -c & \lambda - d \end{vmatrix} = 0$$

Expanding the determinant:

$$\begin{aligned}(\lambda - a)(\lambda - d) - bc &= 0 \\ \lambda^2 - (a + d)\lambda + (ad - bc) &= 0\end{aligned}$$

Solving using the quadratic formula:

$$\begin{aligned}\lambda_{1,2} &= \frac{(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}}{2} \\ \lambda_{1,2} &= \frac{(a + d) \pm \sqrt{(a - d)^2 + 4bc}}{2}\end{aligned}$$

Classification of Critical Points (Origin)

1. **Stable Node:** If λ_1 and λ_2 are real, distinct, and both negative ($\lambda_1 < \lambda_2 < 0$), the system at the origin is a **stable node**.
2. **Unstable Node:** If λ_1 and λ_2 are real, distinct, and both positive ($\lambda_1 > \lambda_2 > 0$), the system at the origin is an **unstable node**.
3. **Stable Center:** If λ_1 and λ_2 are purely imaginary ($\lambda_{1,2} = \pm i\beta$), the system at the origin is a **stable center** (neutrally stable).
4. **Unstable Spiral:** If λ_1 and λ_2 are complex conjugates with positive real part ($\lambda_{1,2} = \alpha \pm i\beta$, $\alpha > 0$), the system at the origin is an **unstable spiral (focus)**.
5. **Stable Spiral:** If λ_1 and λ_2 are complex conjugates with negative real part ($\lambda_{1,2} = \alpha \pm i\beta$, $\alpha < 0$), the system at the origin is a **stable spiral (focus)**.
6. **Unstable Saddle Point:** If one eigenvalue is positive(real) and the other is negative ($\lambda_1 < 0 < \lambda_2$), the system at the origin is an **unstable saddle point**.

3.2 Discuss the stability and nature of the critical point.

$$\begin{aligned}\dot{x} &= x + 3y \\ \dot{y} &= 3x + y\end{aligned}$$

Consider the linear autonomous dynamical system:

$$\begin{aligned}\dot{x} &= x + 3y \\ \dot{y} &= 3x + y\end{aligned}\tag{1}$$

Here, the system matrix coefficients are:

$$a = 1, \quad b = 3, \quad c = 3, \quad d = 1$$

The characteristic equation of system (1) is given by:

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

Substituting the values of a, b, c , and d :

$$\lambda^2 - \lambda(1 + 1) + (1 \cdot 1 - 3 \cdot 3) = 0$$

$$\lambda^2 - 2\lambda - 8 = 0$$

$$\lambda^2 - 4\lambda + 2\lambda - 8 = 0$$

$$\lambda(\lambda - 4) + 2(\lambda - 4) = 0$$

$$(\lambda - 4)(\lambda + 2) = 0$$

Thus, the eigenvalues are:

$$\lambda_1 = 4, \quad \lambda_2 = -2$$

Since one eigenvalue is real and positive ($\lambda_1 = 4 > 0$) and the other eigenvalue is real and negative ($\lambda_2 = -2 < 0$), the critical point at the origin $(0, 0)$ is an **unstable saddle point**.

3.3 Discuss the stability and nature of the critical point of the non linear system and sketch the phase portrait.

$$\begin{aligned}\dot{x} &= x + 4y - x^2 \\ \dot{y} &= 6x - y + 2xy\end{aligned}$$

Solution

Given the non-linear system:

$$\begin{aligned}\dot{x} &= x + 4y - x^2 \\ \dot{y} &= 6x - y + 2xy\end{aligned}\tag{1}$$

Comparing with the standard non-linear system form:

$$\begin{aligned}\dot{x} &= ax + by + P(x, y) \\ \dot{y} &= cx + dy + Q(x, y)\end{aligned}$$

we identify:

$$\begin{aligned}a &= 1, & b &= 4, & c &= 6, & d &= -1 \\ P(x, y) &= -x^2, & Q(x, y) &= 2xy\end{aligned}$$

Clearly, $P(x, y)$ and $Q(x, y)$ are continuous functions. Evaluating the limits as $(x, y) \rightarrow (0, 0)$:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{P(x, y)}{\sqrt{x^2 + y^2}} = \lim_{(x,y) \rightarrow (0,0)} \frac{-x^2}{\sqrt{x^2 + y^2}} = 0$$

and

$$\lim_{(x,y) \rightarrow (0,0)} \frac{Q(x, y)}{\sqrt{x^2 + y^2}} = \lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{\sqrt{x^2 + y^2}} = 0$$

Thus, the condition for the Poincaré–Liapunov linearization theorem holds. The characteristic equation for the corresponding linear system is:

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

Substituting the values $a = 1, b = 4, c = 6, d = -1$:

$$\lambda^2 - \lambda(1 - 1) + [1(-1) - 4(6)] = 0$$

$$\lambda^2 - 0\lambda - 1 - 24 = 0$$

$$\lambda^2 - 25 = 0$$

$$\lambda = \pm 5$$

Since the characteristic roots ($\lambda = 5, -5$) are real and of opposite sign, the critical point $(0, 0)$ of the non-linear system (1) is an ****unstable saddle point****. use the phase portrait for saddle point here.

3.4 Discuss the stability and nature of the critical point of the non linear system and sketch the phase portrait.

$$\begin{aligned}\dot{x} &= \sin x - 4y \\ \dot{y} &= \sin 2x - 5y\end{aligned}$$

Given non-linear system:

$$\begin{aligned}\dot{x} &= \sin x - 4y \\ \dot{y} &= \sin 2x - 5y\end{aligned} \tag{1}$$

Using the Maclaurin series expansion for $\sin x$ and $\sin 2x$:

$$\begin{aligned}\dot{x} &= \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right) - 4y \\ \dot{y} &= \left(2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \dots \right) - 5y\end{aligned}$$

Rearranging into linear and non-linear terms:

$$\begin{aligned}\dot{x} &= x - 4y - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \\ \dot{y} &= 2x - 5y - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \dots\end{aligned}$$

Comparing with $\dot{x} = ax + by + P_1(x, y)$ and $\dot{y} = cx + dy + Q_1(x, y)$:

$$a = 1, \quad b = -4, \quad c = 2, \quad d = -5 \quad (\text{Linear coefficients})$$

$$P_1(x, y) = -\frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$Q_1(x, y) = -\frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \dots$$

Clearly, $P_1(x, y)$ and $Q_1(x, y)$ are polynomial-type series, so they are continuous and differentiable everywhere.

Checking the Poincaré–Liapunov limit conditions:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{P_1(x, y)}{\sqrt{x^2 + y^2}} = 0 \quad \text{and} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{Q_1(x, y)}{\sqrt{x^2 + y^2}} = 0 \quad (\text{Linearization holds})$$

Now, the characteristic equation of the corresponding linear system is:

$$\begin{aligned}\lambda^2 - (a + d)\lambda + (ad - bc) &= 0 & (\det(A - \lambda I) = 0) \\ \lambda^2 - (1 - 5)\lambda + (-5 + 8) &= 0 \\ \lambda^2 + 4\lambda + 3 &= 0 \\ \lambda^2 + 3\lambda + \lambda + 3 &= 0 \\ \lambda(\lambda + 3) + 1(\lambda + 3) &= 0 \\ (\lambda + 1)(\lambda + 3) &= 0 \\ \implies \lambda &= -1, -3 & (\text{Eigenvalues})\end{aligned}$$

Since both roots are real, distinct, and negative ($\lambda_1, \lambda_2 < 0$), the critical point $(0, 0)$ of the system is a **stable node**.

3.5 Find critical point. Discuss the stability and nature of the critical point of the non linear system and sketch the phase portrait.

$$\begin{aligned}\frac{dx}{dt} &= y - x^2 \\ \frac{dy}{dt} &= 27x - y^2\end{aligned}$$

Given non-linear system:

$$\frac{dx}{dt} = y - x^2 \quad (1)$$

$$\frac{dy}{dt} = 27x - y^2 \quad (2)$$

For critical points, set $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$:

$$y - x^2 = 0 \quad (3)$$

$$27x - y^2 = 0 \quad (4)$$

From (3), $y = x^2$. Substituting into (4):

$$\begin{aligned}27x - (x^2)^2 &= 0 \\ x(27 - x^3) &= 0 \implies x = 0, 3\end{aligned}$$

For $x = 0 \implies y = 0$, and for $x = 3 \implies y = 9$. Thus, the critical points are $(0, 0)$ and $(3, 9)$.

For Critical Point $(0, 0)$:

Checking Poincaré–Liapunov limit conditions:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{-x^2}{\sqrt{x^2 + y^2}} = 0 \quad \text{and} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{-y^2}{\sqrt{x^2 + y^2}} = 0 \quad (\text{Linearization holds})$$

The corresponding linearized system is:

$$\frac{dx}{dt} = y, \quad \frac{dy}{dt} = 27x \implies a = 0, b = 1, c = 27, d = 0$$

The characteristic equation is:

$$\lambda^2 - 27 = 0 \implies \lambda = \pm\sqrt{27} = \pm 3\sqrt{3} \quad (\text{Opposite real roots})$$

Since roots are real with opposite signs, the critical point $(0, 0)$ is an **unstable saddle point**.

For Critical Point $(3, 9)$:

Transform coordinates to shift the origin to $(3, 9)$:

$$x = \xi + 3, \quad y = \eta + 9$$

Substituting into (1) and (2):

$$\begin{aligned}\dot{\xi} &= (\eta + 9) - (\xi + 3)^2 = \eta + 9 - \xi^2 - 6\xi - 9 = -6\xi + \eta - \xi^2 \\ \dot{\eta} &= 27(\xi + 3) - (\eta + 9)^2 = 27\xi + 81 - \eta^2 - 18\eta - 81 = 27\xi - 18\eta - \eta^2\end{aligned}\tag{5}$$

Checking limit conditions for non-linear terms $P_1 = -\xi^2$ and $Q_1 = -\eta^2$:

$$\lim_{(\xi, \eta) \rightarrow (0, 0)} \frac{-\xi^2}{\sqrt{\xi^2 + \eta^2}} = 0 \quad \text{and} \quad \lim_{(\xi, \eta) \rightarrow (0, 0)} \frac{-\eta^2}{\sqrt{\xi^2 + \eta^2}} = 0$$

The corresponding linearized system is:

$$\dot{\xi} = -6\xi + \eta, \quad \dot{\eta} = 27\xi - 18\eta \implies a = -6, b = 1, c = 27, d = -18$$

The characteristic equation is:

$$\begin{aligned}\lambda^2 - (-6 - 18)\lambda + (108 - 27) &= 0 \\ \lambda^2 + 24\lambda + 81 &= 0\end{aligned}$$

$$\begin{aligned}\lambda &= \frac{-24 \pm \sqrt{24^2 - 4(1)(81)}}{2} \\ \lambda &= \frac{-24 \pm \sqrt{252}}{2} = -12 \pm \sqrt{63} \quad (\text{Both roots} < 0)\end{aligned}$$

Since both roots are real, distinct, and negative, the critical point $(3, 9)$ is a **stable node**.

Positive Definite:

A function $V(x, y)$ is said to be positive definite in a domain D if $V(0, 0) = 0$ and $V(x, y) > 0$ for all $(x, y) \neq (0, 0)$.

- **Example:** Consider $V(x, y) = x^2 + y^2$, domain $D = \mathbb{R}^2$.

Positive Semi-Definite:

A function $V(x, y)$ is said to be positive semi-definite in a domain D if $V(0, 0) = 0$ and $V(x, y) \geq 0$ for all $(x, y) \neq (0, 0)$.

- **Example:** Consider $V(x, y) = x^2 - y^2$, domain $D = \{(x, y) : x \geq y\}$.

Negative Definite:

A function $V(x, y)$ is said to be negative definite in a domain D if $V(0, 0) = 0$ and $V(x, y) < 0$ for all $(x, y) \neq (0, 0)$.

- **Example:** Consider $V(x, y) = -(x^2 + y^2)$, domain $D = \mathbb{R}^2$.

Negative Semi-Definite:

A function $V(x, y)$ is said to be negative semi-definite in a domain D if $V(0, 0) = 0$ and $V(x, y) \leq 0$ for all $(x, y) \neq (0, 0)$ in D .

- **Example:** Consider $V(x, y) = x^2 - y^2$, domain $D = \{(x, y) : x \leq y\}$.

Liapunov Function:

Suppose that the autonomous system:

$$\begin{aligned}\dot{x} &= P(x, y) \\ \dot{y} &= Q(x, y)\end{aligned}$$

has an isolated critical point $(0, 0)$ and $P(x, y)$ and $Q(x, y)$ have continuous first order partial derivatives for all (x, y) .

If there exists a function $V(x, y)$ related to the system (1) such that:

- $V(x, y)$ is a differentiable function for all x, y .
- $V(x, y)$ is positive definite in a domain D .
- $\dot{V}(x, y) \leq 0$ for all $x, y \in D$.

then $V(x, y)$ is called the **Liapunov Function** of the system (1).

3.6 State the three theorems of Liapunov.17

Theorem 01: Statement: Consider the autonomous system:

$$\begin{aligned}\dot{x} &= P(x, y) \\ \dot{y} &= Q(x, y)\end{aligned}\tag{3.1}$$

If there exists a differentiable function $V(x, y)$ such that $V(x, y)$ is positive definite and $\dot{V}(x, y)$ is negative definite, then the critical point $(0, 0)$ of the system (1) is **asymptotically stable**.

—

Theorem 02: Statement: Consider the autonomous system:

$$\begin{aligned}\dot{x} &= P(x, y) \\ \dot{y} &= Q(x, y)\end{aligned}\tag{3.2}$$

If there exists a differentiable function $V(x, y)$ such that $V(x, y)$ is positive definite and $\dot{V}(x, y)$ is negative semi-definite, then the critical point $(0, 0)$ of the system (1) is **stable**.

—

Theorem 03: Statement: Consider the autonomous system:

$$\begin{aligned}\dot{x} &= P(x, y) \\ \dot{y} &= Q(x, y)\end{aligned}\tag{3.3}$$

If there exists a differentiable function $V(x, y)$ such that $V(x, y)$ is positive definite and $\dot{V}(x, y) > 0$ for all $x \neq 0, y \neq 0$, then the critical point $(0, 0)$ is **unstable**.

3.7 Examine the stability of the solution

$$\begin{aligned}\dot{x} &= -x - y - x^3 \\ \dot{y} &= x - y - y^3\end{aligned} \text{ by taking a suitable Liapunov-} \\ \text{function.2018}$$

Solution:

Given system:

$$\begin{cases} \dot{x} = -x - y - x^3 \\ \dot{y} = x - y - y^3 \end{cases} \quad (3.4)$$

Let $V(x, y) = Ax^m + By^n$ (ii) be the Liapunov function candidate for the system (i), where A and B are positive constants, and m, n are positive even integers.

Taking the total time derivative along the trajectories of system (i):

$$\begin{aligned}\dot{V}(x, y) &= Amx^{m-1}\dot{x} + Bny^{n-1}\dot{y} \\ &= Amx^{m-1}(-x - y - x^3) + Bny^{n-1}(x - y - y^3) \\ &= -Amx^m - Amx^{m-1}y - Amx^{m+2} + Bnxy^{n-1} - Bny^n - Bny^{n+2} \\ &= (-Amx^{m-1}y + Bnxy^{n-1}) - (Amx^m + Amx^{m+2} + Bny^n + Bny^{n+2})\end{aligned}$$

To ensure negative definiteness of $\dot{V}(x, y)$, we set the cross-terms to zero:

$$\begin{aligned}-Amx^{m-1}y + Bnxy^{n-1} &= 0 \\ \implies Amx^{m-1}y &= Bnxy^{n-1}\end{aligned}$$

Comparing coefficients and powers on both sides:

$$Am = Bn \quad \text{and} \quad x^{m-1}y = xy^{n-1} \quad (3.5)$$

Equating exponents:

$$\begin{aligned}m - 1 = 1 &\implies m = 2 \\ n - 1 = 1 &\implies n = 2\end{aligned}$$

Substituting $m = 2$ and $n = 2$ into (iii):

$$2A = 2B \implies A = B$$

Choosing $A = 1$ and $B = 1$:

$$\therefore V(x, y) = x^2 + y^2$$

which is the required Liapunov function.

2nd Part: Stability Analysis

Evaluating $\dot{V}(x, y)$ for $V(x, y) = x^2 + y^2$:

$$\begin{aligned}\dot{V}(x, y) &= 2x\dot{x} + 2y\dot{y} \\ &= 2x(-x - y - x^3) + 2y(x - y - y^3) \\ &= -2x^2 - 2xy - 2x^4 + 2xy - 2y^2 - 2y^4 \\ \therefore \dot{V}(x, y) &= -(2x^2 + 2x^4 + 2y^2 + 2y^4)\end{aligned}$$

1. $V(x, y)$ is continuously differentiable for all $(x, y) \in \mathbb{R}^2$.
2. $V(0, 0) = 0$ and $V(x, y) > 0$ for all $(x, y) \neq (0, 0)$.
 $\therefore V(x, y)$ is **positive definite**.
3. $\dot{V}(x, y) < 0$ for all $(x, y) \neq (0, 0)$.
 $\therefore \dot{V}(x, y)$ is **negative definite**.

Hence, by Liapunov's First Theorem, the critical point $(0, 0)$ is **asymptotically stable**.

3.8 Obtain a Lyapunov function for the dynamical system

$$\dot{x} = 4y - x^3 \quad ((1a))$$

$$\dot{y} = 3x - 4y \quad ((1b))$$

and examine the stability of the origin $(0, 0)$. **Solution:**

Given that:

$$\dot{x} = 4y - x^3 \quad ((i))$$

$$\dot{y} = 3x - 4y$$

Let

$$V(x, y) = Ax^m + By^n \quad ((ii))$$

be the Lyapunov function of the system (i), where A, B are positive constants and m, n are positive and even.

Now differentiating (ii) with respect to t , we get:

$$\begin{aligned} \dot{V}(x, y) &= Amx^{m-1}\dot{x} + Bny^{n-1}\dot{y} \\ &= Amx^{m-1}(4y - x^3) + Bny^{n-1}(3x - 4y) \\ &= 4Amx^{m-1}y - Amx^{m+2} + 3Bny^{n-1}x - 4Bny^n \\ &= 4Amx^{m-1}y + 3Bnxy^{n-1} - (Amx^{m+2} + 4Bny^n) \end{aligned}$$

Since $V(x, y)$ is a Lyapunov function, we put:

$$\begin{aligned} 4Amx^{m-1}y + 3Bnxy^{n-1} &= 0 \\ \Rightarrow 4Amx^{m-1}y &= -3Bnxy^{n-1} \\ \Rightarrow \left. \begin{aligned} 4Am &= -3Bn \\ x^{m-1}y &= xy^{n-1} \end{aligned} \right\} \quad ((iii) \text{ (say)}) \end{aligned}$$

From the second relation of (iii):

$$m - 1 = 1 \implies m = 2$$

$$n - 1 = 1 \implies n = 2$$

From (iii), we get:

$$4A(2) = -3B(2)$$

$$\Rightarrow 4A = -3B$$

$$\Rightarrow \frac{A}{B} = -\frac{3}{4}$$

$$\therefore A = -3, \quad B = 4$$

Thus, the candidate function becomes:

$$\therefore V(x, y) = -3x^2 + 4y^2$$

Differentiating $V(x, y)$ along the system trajectories:

$$\begin{aligned}\therefore \dot{V}(x, y) &= -6x\dot{x} + 8y\dot{y} \\ &= -6x(4y - x^3) + 8y(3x - 4y) \\ &= -24xy + 6x^4 + 24xy - 32y^2 \\ &= 6x^4 - 32y^2\end{aligned}$$

We observe that:

1. $V(x, y)$ is a differentiable function of x and y .
 2. $V(x, y)$ is positive definite for region where $4y^2 > 3x^2$, and $V(x, y) = 0$ if $x = 0, y = 0$.
 3. $\dot{V}(x, y) > 0$ in the region near the origin.
- $\therefore V(x, y)$ is a Lyapunov function for the system (i).
 $\therefore$ The critical point $(0, 0)$ is **unstable**.

Chapter 4

Simple species Population Model

Important Info:

- Malthusian model:

$$\frac{dN}{dt} = kN$$

- solution of M model:

$$N(t) = N_0 e^{kt}$$

- Logistic Model:

$$\frac{dN}{dt} = r \left(1 - \frac{N}{K} \right) N$$

Solution:

$$N = \frac{KN_0}{N_0 + (K - N_0)e^{-rt}}$$

Population

A species which has continuous birth and death rate is called a population.

4.1 Describe and solve the single species population model. Find the doubling/tripling time. Discuss the plausibility and stability conditions.(Malthusian)

The rate of change of population is proportional to the present population. Mathematically:

$$\begin{aligned}\frac{dN}{dt} &\propto N \\ \frac{dN}{dt} &= kN\end{aligned}\tag{1}$$

where k is the proportionality constant (growth/decay rate), which may be positive or negative.

Separating variables and integrating:

$$\begin{aligned}\frac{dN}{N} &= k dt \\ \ln N &= kt + C \\ N &= e^{kt+C} = e^C \cdot e^{kt} \\ N &= Ae^{kt}\end{aligned}\tag{2}$$

where A is constant. Using initial condition $N = N_0$ at $t = 0$:

$$N_0 = Ae^{k(0)} \implies A = N_0$$

Substituting A back into (2):

$$N(t) = N_0 e^{kt}\tag{3}$$

Doubling Time (T_d):

When $t = T_d$, the population doubles, i.e., $N = 2N_0$:

$$\begin{aligned}2N_0 &= N_0 e^{kT_d} \\ e^{kT_d} &= 2 \\ kT_d &= \ln 2 \\ \implies T_d &= \frac{\ln 2}{k} \approx \frac{0.6931}{k}\end{aligned}\tag{Doubling time}$$

Tripling Time (T):

When $t = T$, the population triples, i.e., $N = 3N_0$:

$$\begin{aligned}3N_0 &= N_0 e^{kT} \\ e^{kT} &= 3 \\ kT &= \ln 3 \\ \implies T &= \frac{\ln 3}{k} \approx \frac{1.0986}{k}\end{aligned}\tag{Tripling time}$$

Plausibility Discussion:

- i. For large time $t \rightarrow \infty$ (with $k > 0$), $N(t) \rightarrow \infty$, which gives an absurd/unrealistic result due to limited real-world environmental resources.
- ii. If initial population N_0 is extremely large, the model similarly predicts unrealistic unbounded exponential growth.

We have:

$$N(t) = N_0 e^{kt} \quad (3)$$

From solution (3), we analyze stability based on the parameter k :

1. **Case 1** ($k > 0$):

$$\lim_{t \rightarrow \infty} N(t) \rightarrow \infty$$

In this case, the system is **unstable** due to unbounded growth.

2. **Case 2** ($k = 0$):

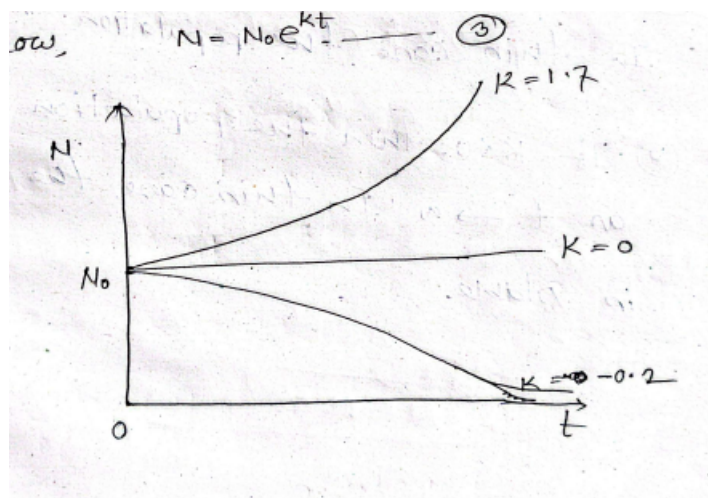
$$N(t) = N_0 \quad \text{for all } t \rightarrow \infty$$

In this case, the population remains constant, so it is **stable** (neutrally stable).

3. **Case 3** ($k < 0$):

$$\lim_{t \rightarrow \infty} N(t) \rightarrow 0$$

In this case, the population is **asymptotically stable** as it decays to zero.



4.2 Assume that, the population of Rajshahi city increases at a rate proportional to the number of inhabitants at any time t . If the population is double in 40 years, how many years it will be triple.

According to the problem, we get:

$$\frac{dN}{dt} = kN \quad (1)$$

The general solution of (1) is:

$$N(t) = N_0 e^{kt} \quad (2)$$

Given that when $t = 40$, $N = 2N_0$:

$$\begin{aligned} 2N_0 &= N_0 e^{40k} \\ 40k &= \ln 2 \\ k &= \frac{\ln 2}{40} \\ \implies k &\approx 0.017328 \end{aligned}$$

Suppose the population will triple in time T , i.e., $N(T) = 3N_0$:

$$\begin{aligned} 3N_0 &= N_0 e^{kT} \\ e^{kT} &= 3 \\ kT &= \ln 3 \\ T &= \frac{\ln 3}{k} \\ T &= \frac{\ln 3}{0.017328} \\ \implies T &\approx 63.400986 \text{ years} \end{aligned}$$

So, the population will triple in 63.400986 years (or approximately 63.40 years from $t = 0$).

4.3 Describe logistic model/Verhulst model. Find its solution. Also find the ultimate/maximum population.

We have the Malthusian model:

$$\frac{dN}{dt} = kN \quad (1)$$

The solution of (1) is:

$$N = N_0 e^{kt} \quad (2)$$

We see that this solution gives accurate results only for small time and population. Therefore, we need to modify model (1). Verhulst assumed that k is a function of population N instead of a constant:

$$\frac{dN}{dt} = f(N)N \quad (3)$$

From reality, when the population is small, it grows rapidly, and after a fixed amount of population, it decreases, i.e.:

$$f(N) = r - aN$$

Therefore, equation (3) becomes:

$$\begin{aligned} \frac{dN}{dt} &= (r - aN)N \\ \Rightarrow \frac{dN}{dt} &= r \left(1 - \frac{aN}{r}\right) N \\ \Rightarrow \frac{dN}{dt} &= r \left(1 - \frac{N}{K}\right) N \end{aligned} \quad (4)$$

where $K = \frac{r}{a}$ is the carrying capacity. Here, equation (4) is called the Verhulst or logistic population model.

2nd Part: Solution of the Logistic Equation

From (4), separating variables:

$$\begin{aligned} \frac{dN}{N \left(1 - \frac{N}{K}\right)} &= r dt \\ \left(\frac{1}{N} + \frac{\frac{1}{K}}{1 - \frac{N}{K}} \right) dN &= r dt \end{aligned}$$

Integrating both sides:

$$\begin{aligned}
\ln N + \frac{1}{K} \cdot \left(-K \ln \left(1 - \frac{N}{K} \right) \right) &= rt + \ln A \\
\ln N - \ln \left(1 - \frac{N}{K} \right) &= rt + \ln A \\
\ln \left(\frac{N}{1 - \frac{N}{K}} \right) &= rt + \ln A \\
\frac{N}{1 - \frac{N}{K}} &= Ae^{rt} \\
\frac{KN}{K - N} &= Ae^{rt} \\
\frac{KN}{K - N} &= Ae^{rt} \tag{5} \\
KN &= A(K - N)e^{rt} \\
KN &= AKe^{rt} - ANe^{rt} \\
N(K + Ae^{rt}) &= AKe^{rt} \\
N &= \frac{AKe^{rt}}{K + Ae^{rt}} \tag{6}
\end{aligned}$$

which is the general solution.

Suppose when $t = 0$, $N = N_0$. Then from equation (5):

$$\frac{KN_0}{K - N_0} = Ae^{r(0)} \implies A = \frac{KN_0}{K - N_0}$$

Substituting A into equation (6):

$$\begin{aligned}
N &= \frac{K \left(\frac{KN_0}{K - N_0} \right) e^{rt}}{K + \left(\frac{KN_0}{K - N_0} \right) e^{rt}} \\
&= \frac{KN_0 e^{rt}}{(K - N_0) + N_0 e^{rt}}
\end{aligned}$$

Dividing numerator and denominator by e^{rt} :

$$N = \frac{KN_0}{N_0 + (K - N_0)e^{-rt}}$$

which is the required particular solution.

3rd Part: Ultimate Solution (Limiting Behavior)

The maximum population (ultimate carrying capacity) as $t \rightarrow \infty$ is:

$$\begin{aligned}
N_{\max} &= \lim_{t \rightarrow \infty} \frac{KN_0}{N_0 + (K - N_0)e^{-rt}} \\
&= \frac{KN_0}{N_0 + (K - N_0)(0)} \\
&= \frac{KN_0}{N_0} = K \\
\implies N_{\max} &= K
\end{aligned}$$

4.4 Find the equilibrium population of logistic model and discuss the stability.

Solution:

We have the logistic model:

$$\begin{aligned}\frac{dN}{dt} &= (r - aN)N \\ &= r \left(1 - \frac{a}{r}N\right) N \\ &= rN \left(1 - \frac{N}{K}\right)\end{aligned}\tag{i}$$

For equilibrium population,

$$\begin{aligned}\frac{dN}{dt} &= 0 \\ rN \left(1 - \frac{N}{K}\right) &= 0 \\ \text{or, } rN &= 0 \quad \text{or} \quad 1 - \frac{N}{K} = 0 \\ \implies N &= 0 \quad \text{or} \quad N = K \quad (\because r \neq 0)\end{aligned}$$

which are the equilibrium populations.

Case i: For $N = 0$

Put $N = 0 + \varepsilon$, where $\varepsilon \ll 1$. Therefore, the model becomes:

$$\begin{aligned}\frac{d(0 + \varepsilon)}{dt} &= r(0 + \varepsilon) \left(1 - \frac{0 + \varepsilon}{K}\right) \\ \implies \frac{d\varepsilon}{dt} &= r\varepsilon \left(1 - \frac{\varepsilon}{K}\right) \\ &= r\varepsilon \quad [\text{Neglecting } \varepsilon^2]\end{aligned}\tag{ii}$$

The solution of (ii) is: $\varepsilon = \varepsilon_0 e^{rt}$ which is an unstable solution.

$\therefore N = 0$ is unstable.

Case ii: For $N = K$

Put $N = K + \varepsilon$, where $\varepsilon \ll 1$.

Therefore, the model becomes:

$$\begin{aligned}\frac{d(K + \varepsilon)}{dt} &= r(K + \varepsilon) \left(1 - \frac{K + \varepsilon}{K}\right) \\ \implies \frac{dK}{dt} + \frac{d\varepsilon}{dt} &= r(K + \varepsilon) \left(1 - 1 - \frac{\varepsilon}{K}\right) \\ \implies 0 + \frac{d\varepsilon}{dt} &= -\frac{r\varepsilon}{K}(K + \varepsilon) \\ \implies \frac{d\varepsilon}{dt} &= -r\varepsilon \quad [\text{Neglecting } \varepsilon^2]\end{aligned}\tag{iii}$$

The solution of (iii) is: $\varepsilon = \varepsilon_0 e^{-rt}$ which decays to 0 as $t \rightarrow \infty$. $\therefore N = K$ is stable.

4.5 Find the doubling time population for logistic model.

We have:

$$\frac{dN}{dt} = r \left(1 - \frac{N}{K}\right) N \quad (1)$$

We know the solution of (1) is:

$$N = \frac{N_0 K}{N_0 + (K - N_0)e^{-rt}} \quad (2)$$

Suppose when t becomes T , the population will double ($N = 2N_0$). Then from (2), we get:

$$\begin{aligned} 2N_0 &= \frac{N_0 K}{N_0 + (K - N_0)e^{-rT}} \\ 2N_0 [N_0 + (K - N_0)e^{-rT}] &= N_0 K \\ 2N_0 + 2(K - N_0)e^{-rT} &= K \\ 2(K - N_0)e^{-rT} &= K - 2N_0 \\ e^{-rT} &= \frac{K - 2N_0}{2(K - N_0)} \\ e^{rT} &= \frac{2(K - N_0)}{K - 2N_0} \\ rT &= \ln \left[\frac{2(K - N_0)}{K - 2N_0} \right] \\ \Rightarrow T &= \frac{1}{r} \ln \left[\frac{2(K - N_0)}{K - 2N_0} \right] \quad (\text{Required doubling time}) \end{aligned}$$

which is the required doubling time.

4.6 A population grows according to the model,

$$\frac{dN}{dt} = 1 - \exp \left[1 - r_0 \left(1 - \frac{N}{K} \right) \right],$$

where r_0 and K are positive constants. Determine the equilibrium state and its stability.

Solution:

We have:

$$\frac{dN}{dt} = 1 - \exp \left[1 - r_0 \left(1 - \frac{N}{K} \right) \right] \quad (1)$$

For equilibrium population, set $\frac{dN}{dt} = 0$:

$$\begin{aligned} 1 - \exp \left[1 - r_0 \left(1 - \frac{N}{K} \right) \right] &= 0 \\ \exp \left[1 - r_0 \left(1 - \frac{N}{K} \right) \right] &= 1 \end{aligned}$$

Taking natural logarithm on both sides:

$$\begin{aligned} 1 - r_0 \left(1 - \frac{N}{K} \right) &= 0 \\ r_0 \left(1 - \frac{N}{K} \right) &= 1 \\ 1 - \frac{N}{K} &= \frac{1}{r_0} \\ \frac{N}{K} &= 1 - \frac{1}{r_0} \\ \implies N &= K \left(1 - \frac{1}{r_0} \right) \quad (\text{Equilibrium population}) \end{aligned}$$

which is the equilibrium population.

Stability Analysis:

Put $N = K \left(1 - \frac{1}{r_0} \right) + \varepsilon$, where $\varepsilon \ll 1$.

Substituting into (1):

$$\begin{aligned} \frac{d}{dt} \left[K \left(1 - \frac{1}{r_0} \right) + \varepsilon \right] &= 1 - \exp \left[1 - r_0 \left(1 - \frac{1}{K} \left[K \left(1 - \frac{1}{r_0} \right) + \varepsilon \right] \right) \right] \\ \frac{d\varepsilon}{dt} &= 1 - \exp \left[1 - r_0 \left(1 - 1 + \frac{1}{r_0} - \frac{\varepsilon}{K} \right) \right] \\ &= 1 - \exp \left(\frac{\varepsilon r_0}{K} \right) \end{aligned}$$

Expanding using Taylor series:

$$\begin{aligned}\frac{d\varepsilon}{dt} &= 1 - \left[1 + \frac{\varepsilon r_0}{K} + \frac{(\varepsilon r_0/K)^2}{2!} + \dots \right] \\ \frac{d\varepsilon}{dt} &= -\frac{\varepsilon r_0}{K} \quad (\text{Neglecting higher-order terms of } \varepsilon) \\ \frac{d\varepsilon}{\varepsilon} &= -\frac{r_0}{K} dt\end{aligned}\tag{2}$$

Integrating (2):

$$\varepsilon(t) = \varepsilon_0 e^{-\frac{r_0}{K}t}$$

As $t \rightarrow \infty$, $\varepsilon(t) \rightarrow 0$.

$$\therefore N^* = K \left(1 - \frac{1}{r_0} \right) \text{ is asymptotically stable.}$$

4.7 The population N of Rajshahi satisfy the Logistic model,

$$\frac{dN}{dt} = 0.03N - 3 \times 10^{-8}N^2,$$

where t is measured in years. If the population of Rajshahi was 2,00,000 in 1980. What will be the population in the year 2000. What would be the possible maximum popⁿ of Rajshahi.

Solution:

We have:

$$\begin{aligned}\frac{dN}{dt} &= 0.03N - 3 \times 10^{-8}N^2 \\ &= 3 \times 10^{-2}N - 3 \times 10^{-8}N^2 \\ &= 3 \times 10^{-2}N (1 - 10^{-6}N) \\ \implies \frac{dN}{N(1 - 10^{-6}N)} &= 3 \times 10^{-2} dt \\ \implies \left(\frac{1}{N} + \frac{10^{-6}}{1 - 10^{-6}N} \right) dN &= 3 \times 10^{-2} dt\end{aligned}\tag{1}$$

Integrating both sides:

$$\begin{aligned}\ln N - \ln (1 - 10^{-6}N) &= 3 \times 10^{-2}t + \ln C \\ \implies \ln \left(\frac{N}{1 - 10^{-6}N} \right) &= \ln (Ce^{3 \times 10^{-2}t}) \\ \implies \frac{N}{1 - 10^{-6}N} &= Ce^{3 \times 10^{-2}t}\end{aligned}\tag{A}$$

When $t = 1980$, $N = 200000$:

$$\begin{aligned}
 C &= \frac{200000 e^{-3 \times 10^{-2} \times 1980}}{1 - 10^{-6} \times 200000} \\
 &= \frac{200000 e^{-59.4}}{0.8} \\
 &= \frac{10^6}{4e^{59.4}}
 \end{aligned}
 \tag{decimal form}$$

Substituting C into equation (A):

$$\begin{aligned}
 \frac{N}{1 - 10^{-6}N} &= \frac{10^6}{4e^{59.4}} \cdot e^{3 \times 10^{-2}t} \\
 \frac{N}{1 - 10^{-6}N} &= \frac{10^6}{4e^{59.4-0.03t}} \\
 4e^{59.4-0.03t}N &= 10^6(1 - 10^{-6}N) \\
 4e^{59.4-0.03t}N &= 10^6 - N \\
 N(1 + 4e^{59.4-0.03t}) &= 10^6 \\
 \implies N(t) &= \frac{10^6}{1 + 4e^{59.4-0.03t}}
 \end{aligned}
 \tag{III}$$

which is the particular solution.

Calculations:

Substituting $t = 2000$ into (III):

$$\begin{aligned}
 N &= \frac{10^6}{1 + 4e^{59.4-0.03 \times 2000}} \\
 &= \frac{10^6}{1 + 4e^{-0.6}} \\
 &= 312964.89 \approx 312965
 \end{aligned}
 \tag{Required population}$$

Maximum population limit:

$$\begin{aligned}
 N_{\max} &= \lim_{t \rightarrow \infty} \frac{10^6}{1 + 4e^{59.4-0.03t}} \\
 &= 10^6
 \end{aligned}$$

4.8 If the carrying capacity of the logistic model is K and $0 < N_0 < \frac{K}{2}$, then determine the time of doubling of population.2018

Solution:

We have the solution of the logistic model:

$$N = \frac{KN_0}{N_0 + (K - N_0)e^{-rt}} \quad (i)$$

Suppose when $t = T$, the population N_0 will double, i.e., $N = 2N_0$.

Putting this in (i), we get:

$$\begin{aligned} 2N_0 &= \frac{KN_0}{N_0 + (K - N_0)e^{-rT}} \\ 2[N_0 + (K - N_0)e^{-rT}] &= K \\ (K - N_0)e^{-rT} &= \frac{K}{2} - N_0 \\ (K - N_0)e^{-rT} &= \frac{K - 2N_0}{2} \\ e^{-rT} &= \frac{K - 2N_0}{2(K - N_0)} \\ e^{rT} &= \frac{2(K - N_0)}{K - 2N_0} \\ rT &= \ln \left[\frac{2(K - N_0)}{K - 2N_0} \right] \\ T &= \frac{1}{r} \ln \left[\frac{2(K - N_0)}{2\left(\frac{K}{2} - N_0\right)} \right] \\ T &= \frac{1}{r} \ln \left[\frac{K - N_0}{\frac{K}{2} - N_0} \right] \end{aligned}$$

Since $0 < N_0 < \frac{K}{2}$, the fraction $\frac{K - N_0}{\frac{K}{2} - N_0}$ is positive (and greater than 1).

$$\therefore \text{The doubling time is } T = \frac{1}{r} \ln \left[\frac{K - N_0}{\frac{K}{2} - N_0} \right]$$

4.9 The world's population was estimated to be 5000 million in 1950 and 7500 million in 2000. Estimate the population of the world in the year 2050, given that the population increases according to the Malthusian model.2023

The Malthusian model of population growth is governed by the differential equation:

$$\frac{dN}{dt} = kN \quad (\text{i})$$

where:

- $N(t)$ is the population (in millions) at time t (in years).
- k is the relative growth rate parameter.

Separating variables in equation (i):

$$\frac{dN}{N} = k dt$$

Integrating both sides:

$$\int \frac{dN}{N} = \int k dt$$

$$\ln N = kt + \ln N_0$$

$$\ln \left(\frac{N}{N_0} \right) = kt$$

$$\implies N(t) = N_0 e^{kt} \quad (\text{ii})$$

—

Let $t = 0$ correspond to the year 1950.

- At $t = 0$ (1950): $N(0) = N_0 = 5000$ million
- At $t = 50$ (2000): $N(50) = 7500$ million
- At $t = 100$ (2050): $N(100) = ?$

From equation (ii), when $t = 50$:

$$N(50) = N_0 e^{50k}$$

$$7500 = 5000 e^{50k}$$

$$\implies e^{50k} = \frac{7500}{5000}$$

$$\implies e^{50k} = 1.5 \quad (\text{iii})$$

For the year 2050, the elapsed time from 1950 is $t = 100$ years:

$$N(100) = N_0 e^{100k}$$

$$N(100) = 5000 e^{100k}$$

Using the laws of exponents:

$$N(100) = 5000 (e^{50k})^2$$

Substituting equation (iii) into this expression:

$$N(100) = 5000 \times (1.5)^2$$

$$N(100) = 5000 \times 2.25$$

$$N(100) = 11250 \text{ million}$$

Ans: The estimated population of the world in the year 2050 is **11,250 million** (or **11.25 billion**).

4.10 Suppose the population of Bangladesh grows according to the Malthusian model. The population of Bangladesh was estimated to be 170.00 million in 2010 and 180.00 million in 2020. Estimate the population of Bangladesh in the year 2030 and also in 2040.

Solution:

The Malthusian model is given by:

$$\frac{dN}{dt} = kN \quad (\text{i})$$

The solution to this model is:

$$N = N_0 e^{k(t-t_0)} \quad (\text{ii})$$

—

1st Case:

Here,

$$N = 180.00 \text{ million}$$

$$N_0 = 170.00 \text{ million}$$

$$t_0 = 2010$$

$$t = 2020$$

From equation (ii):

$$\begin{aligned}180 &= 170 e^{k(2020-2010)} \\ \Rightarrow e^{10k} &= \frac{180}{170} = \frac{18}{17} \\ \Rightarrow k &= \frac{1}{10} \ln \left(\frac{18}{17} \right) \\ &= 0.0057158\end{aligned}$$

Substituting k back into equation (ii):

$$N = N_0 e^{0.0057158(t-t_0)} \quad (\text{iii})$$

—

2nd Case:

Here,

$$\begin{aligned}N_0 &= 180.00 \text{ million} \\ t_0 &= 2020 \\ t &= 2030\end{aligned}$$

From equation (iii), we get:

$$\begin{aligned}N &= 180 e^{0.0057158 \times (2030-2020)} \\ &= 180 e^{0.0057158 \times 10} \\ &= 190 \text{ million}\end{aligned}$$

—

3rd Case:

Here,

$$\begin{aligned}N_0 &= 190 \text{ million} \\ t_0 &= 2030 \\ t &= 2040\end{aligned}$$

Calculating the updated growth rate parameter k :

$$\begin{aligned}k &= \frac{1}{10} \ln \left(\frac{19}{18} \right) \\ &= 5.40672 \times 10^{-3} \\ &= 0.00540672\end{aligned}$$

Estimating population for the year 2040:

$$\begin{aligned}N &= 190 e^{0.00540672 \times (2040-2030)} \\ &= 200.555 \text{ million} \quad (\text{Ans})\end{aligned}$$

4.11 If $\frac{dP}{dt} = rP - k$, $P(0) = P_0$, then solve this equation and make a realistic comment regarding the solution.2020

Solution:

Given differential equation:

$$\frac{dP}{dt} = rP - k, \quad P(0) = P_0$$

Separating variables:

$$\begin{aligned} \frac{dP}{rP - k} &= dt \\ \frac{dP}{r \left(P - \frac{k}{r} \right)} &= dt \\ \implies \frac{dP}{P - \frac{k}{r}} &= r dt \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \int \frac{dP}{P - \frac{k}{r}} &= \int r dt \\ \ln \left(P - \frac{k}{r} \right) &= rt + c \\ P - \frac{k}{r} &= e^{rt+c} \\ P - \frac{k}{r} &= Ae^{rt} \quad \text{where } A = e^c \\ \implies P &= \frac{k}{r} + Ae^{rt} \end{aligned} \tag{i}$$

Applying the initial condition $P(0) = P_0$:

$$\begin{aligned} P_0 &= \frac{k}{r} + Ae^0 \\ \implies A &= P_0 - \frac{k}{r} \end{aligned}$$

Substituting A into equation (i):

$$P(t) = \frac{k}{r} + \left(P_0 - \frac{k}{r} \right) e^{rt} \tag{ii}$$

which is the required solution.

From equation (ii), the dynamics of the population depend on the initial population P_0 relative to the critical threshold $\frac{k}{r}$:

- **Growth Phase** ($P_0 > \frac{k}{r}$): If the initial population P_0 is greater than $\frac{k}{r}$, the term $(P_0 - \frac{k}{r})$ is positive, causing $P(t)$ to grow exponentially over time.
- **Extinction Phase** ($P_0 < \frac{k}{r}$): If the initial population P_0 is less than $\frac{k}{r}$, the harvesting rate k exceeds the natural growth rate rP_0 . Consequently, $P(t)$ decreases over time and eventually reaches zero (extinction).
- **Equilibrium Phase** ($P_0 = \frac{k}{r}$): If $P_0 = \frac{k}{r}$, the population remains perfectly constant at $P(t) = \frac{k}{r}$ for all time $t \geq 0$.

Chapter 5

Discrete population Model

Something Important

- Malthusian Population Model:

$$\frac{dN}{dt} = kN$$

$N(t)$ = population size at time t , k = constant growth rate parameter ($k = b - d$, where b is the per-capita birth rate and d is the per-capita death rate)

solution: $N(t) = N_0 e^{kt}$

- Logistic (Verhulst) Population Model

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right)$$

Its explicit solution is:

$$N(t) = \frac{KN_0}{N_0 + (K - N_0)e^{-rt}}$$

r is the intrinsic growth rate and K is carrying capacity

- Harvesting model:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right) \pm EN$$

Solution:

$$N(t) = \frac{(r \pm E)N_0}{\frac{r}{K}N_0 + \left((r \pm E) - \frac{r}{K}N_0 \right) e^{-(r \pm E)t}}$$

5.1 Describe the Harvesting Population Model and find its solutions. Discuss the stability of the Harvesting Model.

Solution of Harvesting / Migration Model

We have the standard logistic model:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) \quad (\text{i})$$

The mortality rate may be affected by the migration of population from one place to another. The migration is due to better jobs, better weather, better housing, or oppression by the government, etc.

The migration rate is also proportional to the number of population. The logistic model becomes:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) \pm EN \quad (\text{ii})$$

where E is the proportionality constant in the harvesting/migration model, and r, K, E are positive constants. r is called the per capita growth rate, and K is the carrying capacity of the environment.

Solution of the Differential Equation

From ii we have:

$$\begin{aligned} \frac{dN}{dt} &= rN \pm EN - \frac{rN^2}{K} \\ &= N(r \pm E) - \frac{r}{K}N^2 \\ \frac{dN}{dt} &= aN - bN^2 \end{aligned} \quad (\text{A})$$

where $a = r \pm E$ and $b = \frac{r}{K}$.

Separating variables:

$$\begin{aligned} \frac{dN}{N(a - bN)} &= dt \\ \Rightarrow \left(\frac{1}{N} + \frac{b}{a - bN} \right) dN &= a dt \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \ln N - \ln(a - bN) &= at + \ln c \\ \ln \left(\frac{N}{a - bN} \right) &= \ln (ce^{at}) \\ \frac{N}{a - bN} &= ce^{at} \\ N &= (a - bN)ce^{at} \end{aligned} \quad (\text{iii})$$

Applying the initial condition $N(0) = N_0$ at $t = 0$:

$$N_0 = (a - bN_0)c \implies c = \frac{N_0}{a - bN_0}$$

Substituting c back into iii:

$$\begin{aligned} N &= \left(\frac{a - bN}{a - bN_0} \right) N_0 e^{at} \\ N + b \frac{NN_0 e^{at}}{a - bN_0} &= \frac{aN_0 e^{at}}{a - bN_0} \\ N \left(1 + \frac{bN_0 e^{at}}{a - bN_0} \right) &= \frac{aN_0 e^{at}}{a - bN_0} \\ N \left(\frac{a - bN_0 + bN_0 e^{at}}{a - bN_0} \right) &= \frac{aN_0 e^{at}}{a - bN_0} \\ N &= \frac{aN_0 e^{at}}{(a - bN_0) + bN_0 e^{at}} \end{aligned}$$

Dividing numerator and denominator by e^{at} :

$$N(t) = \frac{aN_0}{bN_0 + (a - bN_0)e^{-at}}$$

Substituting back $a = r \pm E$ and $b = \frac{r}{K}$:

$$N(t) = \frac{(r \pm E)N_0}{\frac{r}{K}N_0 + ((r \pm E) - \frac{r}{K}N_0) e^{-(r \pm E)t}}$$

which gives the required population solution.

Stability Analysis

For equilibrium states, set $\frac{dN}{dt} = 0$.

From equation A:

$$\begin{aligned} aN - bN^2 &= 0 \\ N(a - bN) &= 0 \\ \implies N = 0 \quad \text{or} \quad N &= \frac{a}{b} \end{aligned}$$

Case 1: Stability near $N = 0$ Let $N = 0 + \varepsilon$, where ε is a small perturbation.

$$\begin{aligned} \frac{d\varepsilon}{dt} &= r\varepsilon \left(1 - \frac{\varepsilon}{K} \right) \pm E\varepsilon \\ &\approx r\varepsilon \pm E\varepsilon \quad (\text{neglecting } \varepsilon^2) \\ \frac{d\varepsilon}{dt} &= (r \pm E)\varepsilon = a\varepsilon \end{aligned}$$

This is a Malthusian model with solution:

$$\varepsilon(t) = \varepsilon_0 e^{at}$$

For $a > 0$, $\varepsilon(t)$ grows exponentially, so $N = 0$ is an ****unstable**** equilibrium solution.

Case 2: Solution at Equilibrium $N = \frac{a}{b}$ Here $N = \frac{a}{b} = \frac{K}{r}(r \pm E)$.

1. **When $E = 0$ (No harvesting/migration):**

$$N(t) = \frac{rN_0}{\frac{r}{K}N_0 + \left(r - \frac{r}{K}N_0\right)e^{-rt}}$$

which is the standard solution of the logistic population model. The equilibrium population $N = K$ is **stable**.

2. **If $E > r$ (Over-harvesting):** The growth rate becomes negative ($a < 0$). The population will continuously decrease and eventually vanish ($N \rightarrow 0$).
3. **If $E < r$ (Sustainable harvesting):** $a > 0$, so the population stays positive, increases toward the modified carrying capacity, and exists sustainably.

5.2 Determine the extinction time and tripling time population govern by the Model

$$\frac{dN}{dt} = N(4 - N) - 4$$

Population Extinction and Tripling Time

Given that,

$$\begin{aligned} \frac{dN}{dt} &= N(4 - N) - 4 \\ \Rightarrow \frac{dN}{dt} &= 4N - N^2 - 4 \\ \Rightarrow \frac{dN}{dt} &= -(N - 2)^2 \\ \Rightarrow \frac{dN}{(N - 2)^2} &= -dt && \text{(separating variables)} \\ \Rightarrow -(N - 2)^{-1} &= -t - c && \text{(by integrating)} \\ \Rightarrow \frac{1}{N - 2} &= t + c \end{aligned}$$

When $t = 0$, $N = N_0$,

$$\Rightarrow c = \frac{1}{N_0 - 2}$$

$$\begin{aligned} \therefore \frac{1}{N - 2} &= t + \frac{1}{N_0 - 2} \\ \Rightarrow \frac{1}{N - 2} &= \frac{(N_0 - 2)t + 1}{N_0 - 2} \\ \Rightarrow N - 2 &= \frac{N_0 - 2}{(N_0 - 2)t + 1} && \text{(taking reciprocal)} \\ \Rightarrow N &= \frac{2(N_0 - 2)t + 2 + N_0 - 2}{(N_0 - 2)t + 1} \\ \Rightarrow N &= \frac{2(N_0 - 2)t + N_0}{(N_0 - 2)t + 1} \\ \Rightarrow N &= \frac{N_0 + 2N_0t - 4t}{(N_0 - 2)t + 1} \end{aligned}$$

Suppose $t = T$, $N = 0$,

$$\begin{aligned} N_0 + 2N_0T - 4T &= 0 \\ \Rightarrow 4T - 2TN_0 &= N_0 \\ \Rightarrow T &= \frac{N_0}{4 - 2N_0} \\ \Rightarrow T &= \frac{N_0}{2(2 - N_0)}, \quad 0 < N_0 < 2 \end{aligned}$$

which is required time.

Note: Tripling time,

Suppose when $t \rightarrow T$, the population N will be $3N_0$, then

$$\begin{aligned} 3N_0 &= \frac{N_0(2T+1) - 4T}{N_0T - 2T + 1} \\ \Rightarrow 3N_0^2T - 6N_0T + 3N_0 &= 2N_0T + N_0 - 4T && \text{(by cross-multiplying)} \\ \Rightarrow 3N_0^2T - 8N_0T + 2N_0 + 4T &= 0 \\ \Rightarrow 3N_0^2T - 8N_0T + 4T &= -2N_0 \\ \Rightarrow T(3N_0^2 - 8N_0 + 4) &= -2N_0 \\ \therefore T &= \frac{-2N_0}{3N_0^2 - 8N_0 + 4} \end{aligned}$$

5.3 Describe the two species linear competition model and identify it. (not CL)

Let x and y be two competitor species. Then the model is:

$$\frac{dx}{dt} = ax - by \quad (i)$$

$$\frac{dy}{dt} = -cx + dy \quad ; a, b, c, d > 0 \quad (ii)$$

This is called the **linear competition model**.

Identification:

From equation (i), we observe that in the absence of y ($y = 0$), x increases ($\frac{dx}{dt} = ax > 0$), and in the presence of y , the growth rate of x decreases due to the $-by$ term.

Again, from equation (ii), we observe that in the absence of x ($x = 0$), the y population increases ($\frac{dy}{dt} = dy > 0$), and in the presence of x , the y population decreases due to the $-cx$ term.

$\therefore$ Therefore, this is a **two-species linear competition model**.

5.4 Describe the symbiotic population model and explain it. (not CL)

Solution:

Let x and y be two populations. Then the model is:

$$\left. \begin{aligned} \frac{dx}{dt} &= -ax + by \\ \frac{dy}{dt} &= cx - dy \end{aligned} \right\} \quad (i)$$

This is called the **linear symbiotic model**

Here, $a, b, c, d > 0$.

Explanation:

From the 1st equation of (i), we observe that in the absence of y ($y = 0$), x decreases ($\frac{dx}{dt} = -ax < 0$), and in the presence of y , the x population increases due to the $+by$ term.

From the 2nd equation of (i), we observe that in the absence of x ($x = 0$), y decreases ($\frac{dy}{dt} = -dy < 0$), and in the presence of x , the growth rate of y increases due to the $+cx$ term.

$\therefore$ Therefore, this is a **two-species symbiotic model**.

5.5 The population $N(t)$ (in million) of Bangladesh satisfies the model $\frac{dN}{dt} = \alpha N - \beta N^2$ where t is measured in years. If $N(2000) = 127.66$, $N(2010) = 147.56$ and $N(2020) = 164.67$, find the population of Bangladesh in 2030. (2020)

Solution:

Given model:

$$\begin{aligned}\frac{dN}{dt} &= \alpha N - \beta N^2 \\ \Rightarrow \frac{dN}{N(\alpha - \beta N)} &= dt \\ \Rightarrow \frac{1}{\alpha} \left(\frac{1}{N} + \frac{\beta}{\alpha - \beta N} \right) dN &= dt \\ \Rightarrow \ln N - \ln(\alpha - \beta N) &= \alpha t + \ln C \\ \Rightarrow \ln \left(\frac{N}{\alpha - \beta N} \right) &= \ln (C e^{\alpha t}) \\ \Rightarrow \frac{N}{\alpha - \beta N} &= C e^{\alpha t}\end{aligned}$$

Let $A = C$:

$$\frac{N}{\alpha - \beta N} = A e^{\alpha t}$$

Solving for $N(t)$:

$$\begin{aligned}N &= A e^{\alpha t} (\alpha - \beta N) \\ N + \beta A e^{\alpha t} N &= A \alpha e^{\alpha t} \\ N (1 + \beta A e^{\alpha t}) &= A \alpha e^{\alpha t} \\ N &= \frac{A \alpha e^{\alpha t}}{1 + \beta A e^{\alpha t}} = \frac{\alpha}{\beta + \frac{1}{A} e^{-\alpha t}}\end{aligned}$$

Let $c = \frac{1}{A}$:

$$N(t) = \frac{\alpha}{\beta + c e^{-\alpha t}} \quad (i)$$

which is the required solution at any time t .

Given initial/boundary conditions:

- $N(2000) = 127.66$
- $N(2010) = 147.56$
- $N(2020) = 164.67$

From the solution formula:

$$\begin{aligned} N(2000) &= \frac{\alpha}{\beta + ce^{-2000\alpha}} \\ \Rightarrow 127.66 &= \frac{\alpha}{\beta + ce^{-2000\alpha}} \end{aligned} \quad (ii)$$

$$147.56 = \frac{\alpha}{\beta + ce^{-2010\alpha}} \quad (iii)$$

$$164.67 = \frac{\alpha}{\beta + ce^{-2020\alpha}} \quad (iv)$$

Dividing (ii) by (iii):

$$\begin{aligned} \frac{127.66}{147.56} &= \frac{\beta + ce^{-2010\alpha}}{\beta + ce^{-2000\alpha}} \\ \frac{127.66}{147.56} - 1 &= \frac{\beta + ce^{-2010\alpha}}{\beta + ce^{-2000\alpha}} - 1 \\ \frac{127.66 - 147.56}{147.56} &= \frac{c(e^{-2010\alpha} - e^{-2000\alpha})}{\beta + ce^{-2000\alpha}} \\ \frac{-19.90}{147.56} &= \frac{c(e^{-2010\alpha} - e^{-2000\alpha})}{\beta + ce^{-2000\alpha}} \end{aligned} \quad (v)$$

Dividing equation (ii) by (iv):

$$\begin{aligned} \frac{127.66}{164.67} &= \frac{\beta + ce^{-2020\alpha}}{\beta + ce^{-2000\alpha}} \\ \Rightarrow \frac{127.66 - 164.67}{164.67} &= \frac{c(e^{-2020\alpha} - e^{-2000\alpha})}{\beta + ce^{-2000\alpha}} \\ \Rightarrow \frac{-37.01}{164.67} &= \frac{c(e^{-2020\alpha} - e^{-2000\alpha})}{\beta + ce^{-2000\alpha}} \end{aligned} \quad (vii)$$

Dividing (v) by (vii):

$$\begin{aligned} \left(\frac{-19.90}{147.56} \right) \times \left(\frac{164.67}{-37.01} \right) &= \frac{e^{-2010\alpha} - e^{-2000\alpha}}{e^{-2020\alpha} - e^{-2000\alpha}} \\ \Rightarrow 0.600039 &= \frac{e^{-2000\alpha}(e^{-10\alpha} - 1)}{e^{-2000\alpha}(e^{-20\alpha} - 1)} \\ \Rightarrow 0.600039 &= \frac{e^{-10\alpha} - 1}{e^{-20\alpha} - 1} = \frac{e^{-10\alpha} - 1}{(e^{-10\alpha})^2 - 1} \\ \Rightarrow 0.600039 &= \frac{1}{e^{-10\alpha} + 1} \\ \Rightarrow e^{-10\alpha} + 1 &= 1.666558 \\ \Rightarrow e^{-10\alpha} &= 0.666558 \\ \Rightarrow \alpha &= -\frac{1}{10} \ln(0.666558) \\ \Rightarrow \alpha &= 0.0405628 \end{aligned}$$

—

From equations (iii) and (iv):

$$\beta + ce^{-2010\alpha} = \frac{\alpha}{147.56 \times 10^6} \quad (\text{viii})$$

$$\beta + ce^{-2020\alpha} = \frac{\alpha}{164.67 \times 10^6} \quad (\text{ix})$$

Subtracting (ix) - (viii):

$$\begin{aligned} c(e^{-2020\alpha} - e^{-2010\alpha}) &= \frac{\alpha}{10^6} \left(\frac{1}{164.67} - \frac{1}{147.56} \right) \\ \Rightarrow c &= \frac{0.0405628}{10^6 \times (e^{-2020\alpha} - e^{-2010\alpha})} \left(\frac{1}{164.67} - \frac{1}{147.56} \right) \\ \Rightarrow c &= 2.19455 \times 10^{25} \end{aligned}$$

From equation (viii):

$$\begin{aligned} \beta + 2.19455 \times 10^{25} \times e^{-2010 \times 0.0405628} &= \frac{0.0405628}{147.56 \times 10^6} \\ \Rightarrow \beta + 8.56609 \times 10^{-11} &= 2.74890 \times 10^{-10} \\ \Rightarrow \beta &= 2.74890 \times 10^{-10} - 8.56609 \times 10^{-11} \\ \Rightarrow \beta &= 1.892283 \times 10^{-10} \end{aligned}$$

When $t = 2030$:

$$\begin{aligned} N(2030) &= \frac{0.0405628}{1.892283 \times 10^{-10} + 2.19455 \times 10^{25} e^{-0.0405628 \times 2030}} \\ &= \frac{0.0405628}{2.272884 \times 10^{-10}} \\ &\approx 178463981.7 \end{aligned}$$

$$\mathbf{N(2030) \approx 178.46 \text{ million}}$$

Chapter 6

Two species Population Model

Important Info:

- linear predator prey model

$$\frac{dx}{dt} = -ax + by \tag{1}$$

$$\frac{dy}{dt} = dy - cx ; \quad a, b, c, d > 0 \tag{2}$$

6.1 Describe the two species linear predator prey model and explain it.

Solution: Let x denote the predator population and y denote the prey population. Then the model:

$$\frac{dx}{dt} = -ax + by \quad (1)$$

$$\frac{dy}{dt} = dy - cx ; \quad a, b, c, d > 0 \quad (2)$$

is called the two-species linear predator-prey model.

2nd part / (Explanation):

From equation (1), we observe that in the absence of prey ($y = 0$), $\frac{dx}{dt} = -ax$, meaning the predator population x decays exponentially. When prey is present ($by > 0$), it contributes positively to predator growth.

From equation (2), we observe that in the absence of predators ($x = 0$), $\frac{dy}{dt} = dy$, meaning the prey population y grows exponentially (Malthusian growth). When predators are present ($-cx < 0$), predation reduces prey population growth.

$\therefore$ The model is a two-species linear predator-prey model.

6.2 Identify and solve the two species linear population model.

Solution: Given that

$$\frac{dx}{dt} = -x + 4y \quad (\text{i})$$

$$\frac{dy}{dt} = 5y - 2x \quad (\text{ii})$$

From equation (i), we observe that in absence of y , the x population decreases and in presence of y , the x population increases.

Again, from equation (ii), we observe that in absence of x , the y population increases and in presence of x , the y population decreases.

$\therefore$ The model is a predator–prey population model.

2nd part: The model can be written as

$$(D + 1)x - 4y = 0 \quad (\text{iii})$$

$$2x + (D - 5)y = 0 \quad (\text{iv})$$

where $D = \frac{d}{dt}$.

Now, multiplying (iii) by $(D - 5)$ and (iv) by 4 and adding:

$$\begin{array}{rcl} (D + 1)(D - 5)x - 4(D - 5)y & = & 0 \\ 8x + 4(D - 5)y & = & 0 \\ \hline (D + 1)(D - 5)x + 8x & = & 0 \\ \Rightarrow (D^2 - 4D - 5)x + 8x & = & 0 \\ \Rightarrow (D^2 - 4D - 5 + 8)x & = & 0 \\ \Rightarrow (D^2 - 4D + 3)x & = & 0 \end{array} \quad (\text{v})$$

The trial solution of (v) is $x = ce^{mt}$.

$$\begin{aligned} \therefore \text{The Auxiliary equation } m^2 - 4m + 3 &= 0 \\ \Rightarrow (m - 3)(m - 1) &= 0 \\ \Rightarrow m &= 1, 3 \end{aligned}$$

$$\therefore \text{The general solution is } x = c_1e^t + c_2e^{3t} \quad (\text{vi})$$

Putting the value in (iii):

$$\begin{aligned} (D + 1)(c_1e^t + c_2e^{3t}) &= 4y \\ \Rightarrow 4y &= D(c_1e^t + c_2e^{3t}) + c_1e^t + c_2e^{3t} \\ \Rightarrow 4y &= c_1e^t + 3c_2e^{3t} + c_1e^t + c_2e^{3t} \\ \Rightarrow y &= \frac{1}{2}c_1e^t + c_2e^{3t} \end{aligned} \quad (\text{Answer})$$

6.3 State and explain the Lotka-Volterra predator prey model

Also find the exact solution and make a stability analysis and sketch the phase portrait of the population dynamics

Statement and Explanation of the Model

Let $x(t)$ denote the prey population density and $y(t)$ denote the predator population density at time t . The classic Lotka–Volterra system is defined by:

$$\frac{dx}{dt} = \alpha x - \beta xy \quad (1)$$

$$\frac{dy}{dt} = -\gamma y + \delta xy \quad (2)$$

where parameters $\alpha, \beta, \gamma, \delta > 0$.

- αx : Exponential growth of prey in the absence of predators.
- $-\beta xy$: Rate of prey mortality due to predation (proportional to interaction frequency).
- $-\gamma y$: Exponential death/decay rate of predators in the absence of prey.
- δxy : Rate of predator population increase resulting from consuming prey.

Explanation:

From the prey equation, we observe that in the absence of predators, the prey population increases without limit, and in the presence of predators, the growth rate of the prey population is reduced by an amount proportional to the predator population present.

Similarly, from the predator equation, we find that in the absence of prey, the predator population decreases and becomes extinct due to starvation; and in the presence of the prey population, the growth rate of the predator population is increased by an amount proportional to the prey population present.

Exact Solution

The model (1) can be written as:

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{y(-\gamma + \delta x)}{x(\alpha - \beta y)} \\
 \Rightarrow \frac{dy}{dx} &= \frac{xy \left(-\frac{\gamma}{x} + \delta\right)}{xy \left(\frac{\alpha}{y} - \beta\right)} \\
 \Rightarrow \frac{dy}{dx} &= \frac{\delta - \frac{\gamma}{x}}{\frac{\alpha}{y} - \beta} \\
 \Rightarrow \left(\frac{\alpha}{y} - \beta\right) dy &= \left(\delta - \frac{\gamma}{x}\right) dx && \text{(separating variables)} \\
 \Rightarrow \alpha \frac{dy}{y} - \beta dy &= \delta dx - \gamma \frac{dx}{x}
 \end{aligned}$$

Integrating, we get:

$$\begin{aligned}
 \alpha \ln y - \beta y &= \delta x - \gamma \ln x + \ln k && \text{(by integrating)} \\
 \Rightarrow \ln y^\alpha - \ln e^{\beta y} &= \ln e^{\delta x} - \ln x^\gamma + \ln k \\
 \Rightarrow \ln y^\alpha - \ln e^{\beta y} - \ln e^{\delta x} + \ln x^\gamma &= \ln k \\
 \Rightarrow \ln \frac{y^\alpha}{e^{\beta y}} + \ln \frac{x^\gamma}{e^{\delta x}} &= \ln k \\
 \Rightarrow \ln \frac{x^\gamma y^\alpha}{e^{\delta x} e^{\beta y}} &= \ln k \\
 \Rightarrow \frac{x^\gamma y^\alpha}{e^{\delta x + \beta y}} &= k && \text{(II)}
 \end{aligned}$$

where k is the integral constant, which is the **general solution**.

Using initial conditions $x(0) = x_0$ and $y(0) = y_0$, we get:

$$\frac{x_0^\gamma y_0^\alpha}{e^{\delta x_0} e^{\beta y_0}} = k$$

Putting the value of k in (II), we get:

$$\frac{x^\gamma y^\alpha}{e^{\delta x} e^{\beta y}} = \frac{x_0^\gamma y_0^\alpha}{e^{\delta x_0} e^{\beta y_0}} \quad \text{(III)}$$

which is the **particular solution** of (1).

Using $f(x) = \frac{x^\gamma}{e^{\delta x}}$ and $g(y) = \frac{y^\alpha}{e^{\beta y}}$:

$$\therefore f(x) \cdot g(y) = K \quad \text{(IV)}$$

Extrema Analysis for $f(x)$

From (IV), it is observed that $f(x)$ and $g(y)$ are inversely related.

Also, $f(0) = 0$ when $x = 0$, and $f(x) \rightarrow 0$ as $x \rightarrow \infty$.

Now,

$$\frac{df}{dx} = (\gamma - \delta x)x^{\gamma-1}e^{-\delta x}$$

and

$$\frac{d^2f}{dx^2} = [(\gamma - \delta x)^2 - \gamma] x^{\gamma-2}e^{-\delta x}$$

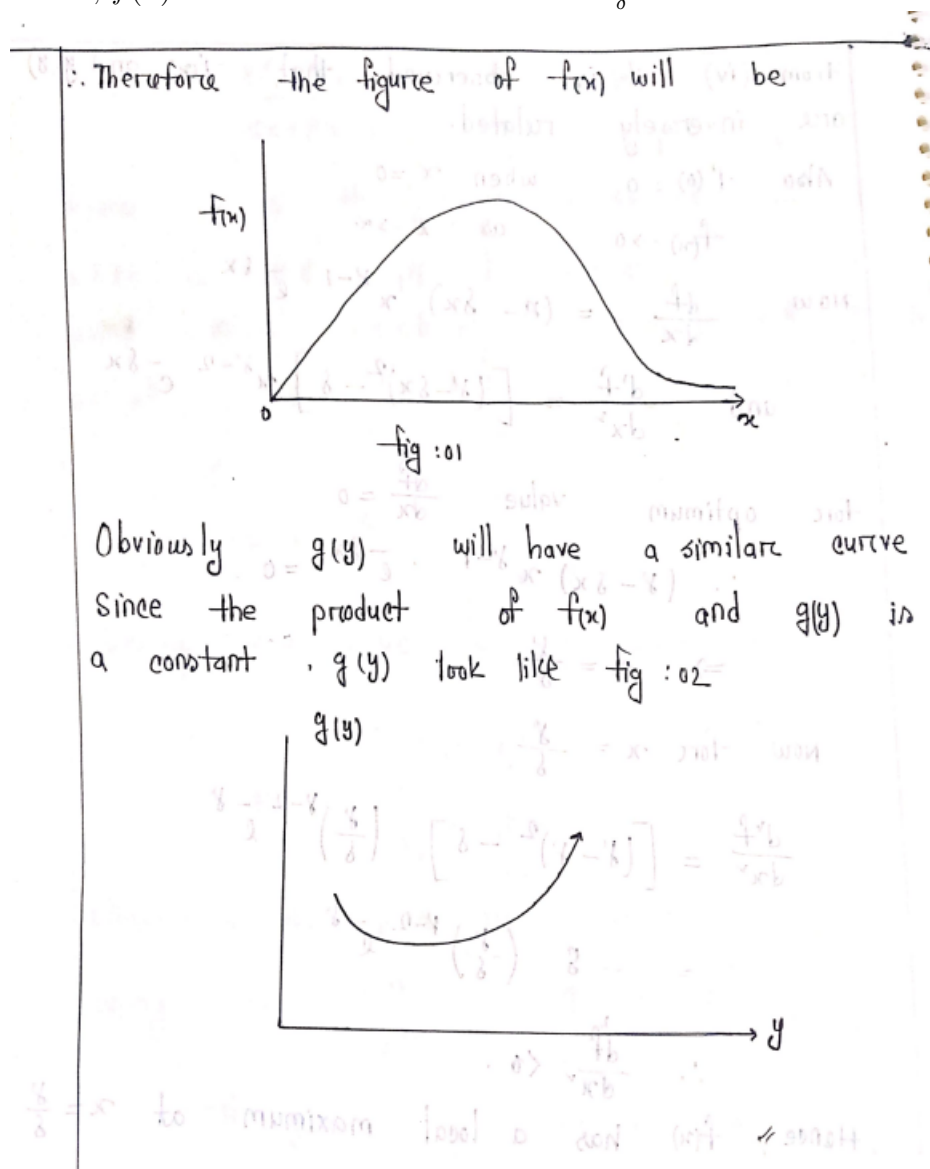
For optimum value, $\frac{df}{dx} = 0$:

$$\therefore (\gamma - \delta x)x^{\gamma-1}e^{-\delta x} = 0 \implies x = \frac{\gamma}{\delta}$$

Now for $x = \frac{\gamma}{\delta}$:

$$\begin{aligned} \frac{d^2f}{dx^2} &= [(\gamma - \gamma)^2 - \gamma] \left(\frac{\gamma}{\delta}\right)^{\gamma-2} e^{-\gamma} = -\gamma \left(\frac{\gamma}{\delta}\right)^{\gamma-2} e^{-\gamma} \\ \therefore \frac{d^2f}{dx^2} &< 0 \end{aligned}$$

Hence, $f(x)$ has a **local maximum** at $x = \frac{\gamma}{\delta}$.



Stability Analysis

System (i) cannot be solved to find $x(t)$ and $y(t)$ explicitly, so the exact path of (i) cannot be drawn. We look for points where the population remains constant (for an indefinite time) and find the paths of (i) in the neighborhood of these equilibrium points.

The equilibrium points of (i) are given by:

$$\begin{cases} x(\alpha - \beta y) = 0 \\ y(-\gamma + \delta x) = 0 \end{cases} \quad (v)$$

Solving (v), we find the equilibrium points $A(0,0)$ and $B\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$.

We note that these are also solutions of (i) which are independent of time and can be considered as special cases in which the curves degenerate into points.

We now investigate the stability of $A(0,0)$ and $B\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$.

Case I — Equilibrium Point $A(0,0)$:

This is the case where both the prey and the predator populations are absent. Thus, when the populations are sufficiently close to this point, we can neglect the second terms βxy and δxy of (i) in comparison to the first terms. Hence, (i) reduces to:

$$\begin{cases} \frac{dx}{dt} = \alpha x \\ \frac{dy}{dt} = -\gamma y \end{cases} \quad (vi)$$

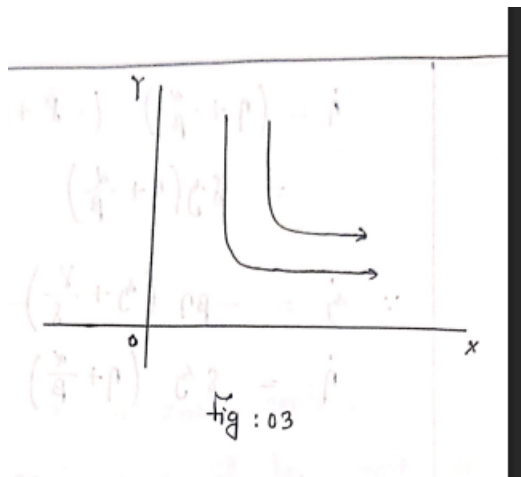
with solutions:

$$x(t) = c_1 e^{\alpha t} \quad \text{and} \quad y(t) = c_2 e^{-\gamma t} \quad (vii)$$

where c_1 and c_2 are arbitrary constants. Since x and y are non-negative, we must have $c_1 \geq 0$, $c_2 \geq 0$, and the family of curves described by (vii) is shown in **fig: 03**.

Further, if we displace the population slightly from the equilibrium point $(0,0)$, then it tends to move away from the point $(0,0)$.

$\therefore$ The equilibrium point $A(0,0)$ is **unstable**.



Case II — Equilibrium Point $B\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$ Perturbation

This is the case where both species are present. We consider (i) by using:

$$\left. \begin{aligned} x &= \xi + \frac{\gamma}{\delta} \\ y &= \eta + \frac{\alpha}{\beta} \end{aligned} \right\} \quad (\text{ix})$$

$$\begin{aligned} \frac{d\xi}{dt} &= \left(\xi + \frac{\gamma}{\delta}\right)(\alpha - \beta\eta - \alpha) = -\beta\eta\left(\xi + \frac{\gamma}{\delta}\right) \\ \therefore \dot{\xi} &= -\beta\eta\left(\xi + \frac{\gamma}{\delta}\right) \\ \implies \dot{\xi} &= -\beta\xi\eta - \frac{\gamma\beta}{\delta}\eta \\ \therefore \text{Similarly } \dot{\eta} &= \delta\xi\eta + \frac{\alpha\delta}{\beta}\xi \end{aligned}$$

Since ξ and η are small, we can neglect the non-linear terms of $\xi\eta$:

$$\left. \begin{aligned} \dot{\xi} &= -\frac{\gamma\beta}{\delta}\eta = a\xi + b\eta \\ \dot{\eta} &= \frac{\alpha\delta}{\beta}\xi = c\xi + d\eta \end{aligned} \right\} \quad (\text{x})$$

Here $a = 0, b = -\frac{\gamma\beta}{\delta}, c = \frac{\alpha\delta}{\beta}, d = 0$. (linearized equation)

$\therefore$ The characteristic equation is:

$$\begin{aligned} \lambda^2 - (a + d)\lambda + (ad - bc) &= 0 \\ \implies \lambda^2 - (0 + 0) + \left(0 + \frac{\gamma\beta}{\delta} \cdot \frac{\alpha\delta}{\beta}\right) &= 0 \\ \implies \lambda^2 &= -\alpha\gamma \\ \implies \lambda &= \pm\sqrt{-\alpha\gamma} \\ \therefore \lambda &= \pm i\sqrt{\alpha\gamma} \end{aligned}$$

Thus, $B\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$ is a **stable center** (Eigen value is purely imaginary). To know the exact shape of the center, we eliminate t from (x) and obtain:

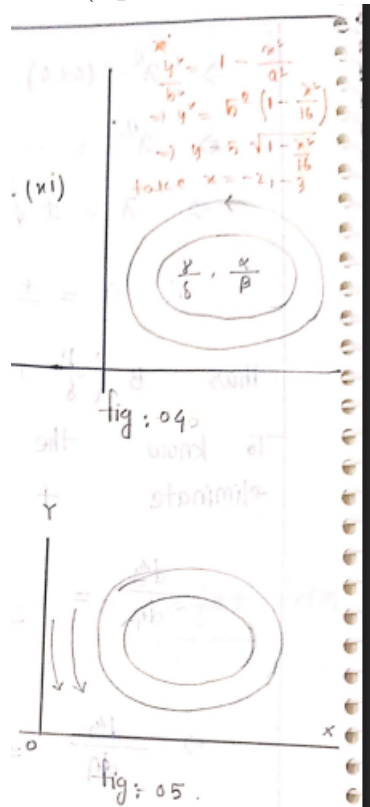
$$\begin{aligned} \frac{d\xi}{d\eta} &= \frac{-\frac{\gamma\beta}{\delta}\eta}{\frac{\alpha\delta}{\beta}\xi} = -\frac{\gamma\beta}{\delta}\eta \times \frac{\beta}{\alpha\delta\xi} = -\frac{\gamma\beta^2}{\alpha\delta^2} \frac{\eta}{\xi} \\ \implies \xi d\xi &= -\frac{\gamma\beta^2}{\alpha\delta^2} \eta d\eta \\ \implies \frac{1}{\gamma\beta^2} \xi d\xi + \frac{1}{\alpha\delta^2} \eta d\eta &= 0 \end{aligned}$$

Integrating, we get:

$$\begin{aligned}
 \frac{1}{\gamma\beta^2} \frac{\xi^2}{2} + \frac{1}{\alpha\delta^2} \frac{\eta^2}{2} &= \frac{c^2}{2} \\
 \Rightarrow \frac{\xi^2}{\gamma\beta^2} + \frac{\eta^2}{\alpha\delta^2} &= c^2 \\
 \Rightarrow \frac{\xi^2}{\gamma\beta^2 c^2} + \frac{\eta^2}{\alpha\delta^2 c^2} &= 1
 \end{aligned} \tag{xi}$$

where c is a constant.

This represents a family of **concentric ellipses**. We find from (xi) that the directions of the ellipses are **anti-clockwise**. (eqn x has different sign) Hence, the stable center



$B\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$ is shown in **fig: 04**.

From **fig: 01**, it is clear that for any value of $f(x)$, there are either two values of x or none. Same arguments hold for y also. This implies that the path of (i) is a **closed loop** as shown in **fig: 05**.

Hence, from **fig: 03**, **fig: 04**, and **fig: 05**, the population dynamics of (i) near its equilibrium points are shown in the phase plane.

6.4 Problem: Two-Species Competition Model 2021

Identify and solve the two-species linear population model:

$$\frac{dx}{dt} = 3x - y \quad (x(0) = 95)$$

$$\frac{dy}{dt} = -2x + 2y \quad (y(0) = 165)$$

Determine the time when the population will become extinct.

1st Part: Model Identification

Consider the system of differential equations:

$$\frac{dx}{dt} = 3x - y \quad (1a)$$

$$\frac{dy}{dt} = -2x + 2y \quad (1b)$$

From equation (1a), in the absence of y , x increases, whereas in the presence of y , x decreases. From equation (1b), in the absence of x , y increases, whereas in the presence of x , y decreases.

Since both species inhibit each other's growth rate, system (1) is a **competition model**.

2nd Part: General Solution

Using the differential operator $D = \frac{d}{dt}$:

$$(D - 3)x + y = 0 \quad (ii)$$

$$2x + (D - 2)y = 0 \quad (iii)$$

Applying $[(ii) \times (D - 2) - (iii)]$:

$$\begin{aligned} (D - 3)(D - 2)x + (D - 2)y - 2x - (D - 2)y &= 0 \\ (D^2 - 5D + 6)x - 2x &= 0 & \text{(eliminating } y) \\ (D^2 - 5D + 4)x &= 0 & (iv) \end{aligned}$$

Equation (iv) is a homogeneous linear 2nd-order differential equation. Let $x = e^{mt}$ be a trial solution:

$$\begin{aligned} m^2 - 5m + 4 &= 0 & \text{(since } e^{mt} \neq 0) \\ (m - 1)(m - 4) &= 0 \implies m = 1, 4 \end{aligned}$$

Hence, the general solution for $x(t)$ is:

$$x(t) = c_1 e^t + c_2 e^{4t}$$

Expressing $y(t)$ using equation (1a):

$$\begin{aligned} y &= 3x - \frac{dx}{dt} & \text{(from (1a))} \\ y &= 3(c_1 e^t + c_2 e^{4t}) - \frac{d}{dt}(c_1 e^t + c_2 e^{4t}) \\ y &= 3c_1 e^t + 3c_2 e^{4t} - (c_1 e^t + 4c_2 e^{4t}) \\ \implies y(t) &= 2c_1 e^t - c_2 e^{4t} \end{aligned}$$

Thus, the general solution of the model is:

$$\begin{aligned}x(t) &= c_1 e^t + c_2 e^{4t} \\y(t) &= 2c_1 e^t - c_2 e^{4t}\end{aligned}\tag{v}$$

3rd Part: Particular Solution & Extinction Analysis

Applying initial conditions $x(0) = 95$ and $y(0) = 165$:

$$\begin{aligned}95 &= c_1 + c_2 \\165 &= 2c_1 - c_2\end{aligned}$$

Adding both equations gives $3c_1 = 260 \implies c_1 = \frac{260}{3}$, which yields $c_2 = 95 - \frac{260}{3} = \frac{25}{3}$.
The particular solution is:

$$x(t) = \frac{260}{3}e^t + \frac{25}{3}e^{4t}\tag{vi}$$

$$y(t) = \frac{520}{3}e^t - \frac{25}{3}e^{4t}\tag{vii}$$

Extinction Time Determination:

Let $t = T$ be the extinction time where population reaches zero.

For species x :

$$\begin{aligned}0 &= \frac{260}{3}e^T + \frac{25}{3}e^{4T} && (\text{set } x(T) = 0) \\e^{3T} &= -\frac{260}{25} \implies T = \frac{1}{3} \ln \left(-\frac{260}{25} \right)\end{aligned}$$

Since $\ln(-260/25) \notin \mathbb{R}$, population x will **never become extinct**.

For species y :

$$\begin{aligned}0 &= \frac{520}{3}e^T - \frac{25}{3}e^{4T} && (\text{set } y(T) = 0) \\e^{3T} &= \frac{520}{25} = \frac{104}{5} \\ \implies T &= \frac{1}{3} \ln \left(\frac{104}{5} \right) \approx 1.0118\end{aligned}$$

Since $T > 0$ is real and positive, species y will become **extinct at time** $T = \frac{1}{3} \ln \left(\frac{104}{5} \right)$.

6.5 Identify and solve the two-species linear population model:

$$\begin{aligned}\frac{dx}{dt} &= -2x + 4y & (x(0) = 100) \\ \frac{dy}{dt} &= x - 2y & (y(0) = 300)\end{aligned}$$

Also find the time when the population will become extinct.2020

Solution:

1st Part: Model Identification

Given system of differential equations:

$$\begin{aligned}\frac{dx}{dt} &= -2x + 4y & (i) \\ \frac{dy}{dt} &= x - 2y & (ii)\end{aligned}$$

From equation (i), in the absence of y , the growth rate of x decreases, while in the presence of y , the growth rate of x increases. Similarly, from equation (ii), in the absence of x , the growth rate of y decreases, while in the presence of x , the growth rate of y increases.

Therefore, the system represents a **symbiotic model**.

2nd Part: General Solution

Using the differential operator $D = \frac{d}{dt}$:

$$\begin{aligned}(D + 2)x - 4y &= 0 & (iii) \\ -x + (D + 2)y &= 0 & (iv)\end{aligned}$$

Operating $(D + 2)$ on (iii) and multiplying (iv) by 4:

$$\begin{aligned}(D + 2)^2 x - 4(D + 2)y &= 0 \\ -4x + 4(D + 2)y &= 0\end{aligned}$$

Adding both equations eliminates y :

$$\begin{aligned}[(D + 2)^2 - 4]x &= 0 & (\text{eliminating } y) \\ (D^2 + 4D)x &= 0 \\ \implies \frac{d^2 x}{dt^2} + 4\frac{dx}{dt} &= 0 & (v)\end{aligned}$$

Let $x = e^{mt}$ be a trial solution of (v). The auxiliary equation is:

$$\begin{aligned}m^2 + 4m &= 0 \\ m(m + 4) &= 0 \implies m = 0, -4\end{aligned}$$

Hence, the general solution for $x(t)$ is:

$$x(t) = c_1 + c_2 e^{-4t} \quad (\text{vi})$$

Substituting equation (vi) into equation (i) to solve for $y(t)$:

$$\begin{aligned} 4y &= \frac{dx}{dt} + 2x && (\text{from (i)}) \\ 4y &= -4c_2 e^{-4t} + 2(c_1 + c_2 e^{-4t}) \\ 4y &= 2c_1 - 2c_2 e^{-4t} \\ \implies y(t) &= \frac{1}{2}c_1 - \frac{1}{2}c_2 e^{-4t} && (\text{vii}) \end{aligned}$$

Equations (vi) and (vii) are the general solution.

3rd Part: Particular Solution

Applying initial conditions $x(0) = 100$ and $y(0) = 300$:

$$100 = c_1 + c_2 \quad (\text{viii})$$

$$300 = \frac{1}{2}c_1 - \frac{1}{2}c_2 \quad (\text{ix})$$

From equation (viii), $c_1 = 100 - c_2$. Substituting into (ix):

$$\begin{aligned} 300 &= \frac{1}{2}(100 - c_2) - \frac{1}{2}c_2 \\ 300 &= 50 - c_2 \implies c_2 = -250 \\ \implies c_1 &= 100 - (-250) = 350 \end{aligned}$$

Substituting c_1 and c_2 gives the particular solution:

$$x(t) = 350 - 250e^{-4t} \quad (\text{x})$$

$$y(t) = 175 + 125e^{-4t} \quad (\text{xi})$$

4th Part: Extinction Time Analysis

Let $t = T$ be the extinction time where population reaches 0.

For species x :

$$\begin{aligned} 0 &= 350 - 250e^{-4T} && (\text{xii}) \\ 250e^{-4T} &= 350 \\ e^{-4T} &= \frac{35}{25} = \frac{7}{5} \\ \ln(e^{-4T}) &= \ln\left(\frac{7}{5}\right) \\ -4T &= \ln\left(\frac{7}{5}\right) \implies T = -\frac{1}{4}\ln\left(\frac{7}{5}\right) \end{aligned}$$

Since time cannot be negative ($T < 0$), the x population will **never go extinct**.

For species y :

$$\begin{aligned} 0 &= 175 + 125e^{-4T} & (\text{xiii}) \\ e^{-4T} &= -\frac{175}{125} = -\frac{7}{5} \\ -4T &= \ln\left(-\frac{7}{5}\right) \implies T = -\frac{1}{4}\ln\left(-\frac{7}{5}\right) \end{aligned}$$

Since $\ln(-7/5)$ is undefined in the real numbers ($\mathbb{R}$), the y population will **never go extinct**.

Check page 146, 2 math

Chapter 7

Epidemic/Infection disease Model

Info:

- SIR Model:

$$\frac{dS}{dt} = -\beta SI \quad (1)$$

$$\frac{dI}{dt} = \beta SI - \gamma I \quad (2)$$

$$\frac{dR}{dt} = \gamma I \quad (3)$$

7.1 Describe the simple SIR epidemic model. Find the condition on which the infection will ultimately die out or spread throughout the population.

Description of the Simple SIR Epidemic Model

The **SIR model** (Kermack and McKendrick, 1927) divides a fixed total population N into three compartmental classes:

- $S(t)$: **Susceptible** individuals who are healthy but vulnerable to catching the disease.
- $I(t)$: **Infected** individuals who currently have the disease and can transmit it to susceptibles.
- $R(t)$: **Removed** (or Recovered) individuals who have either recovered with permanent immunity or died.

Model Assumptions:

1. The total population N remains constant over the epidemic time scale (i.e., no births, natural deaths, or migration):

$$S(t) + I(t) + R(t) = N = \text{constant}$$

2. Infection rate is proportional to the rate of contact between susceptible and infected individuals (βSI).
3. Recovery rate is directly proportional to the number of infected individuals (γI).

Differential Equations:

$$\frac{dS}{dt} = -\beta SI \tag{1}$$

$$\frac{dI}{dt} = \beta SI - \gamma I \tag{2}$$

$$\frac{dR}{dt} = \gamma I \tag{3}$$

where:

- $\beta > 0$ is the **transmission (infection) rate coefficient**.
- $\gamma > 0$ is the **recovery/removal rate coefficient**.

The initial conditions of the above model are:

$$S(0) = S_0 > 0, \quad I(0) = I_0 > 0, \quad \text{and} \quad R(0) = 0$$

Initially, from equation (2):

$$\begin{aligned}\frac{dI}{dt} &= \beta SI - \alpha I \\ \implies \frac{dI}{dt} &= I\beta \left(S - \frac{\alpha}{\beta} \right)\end{aligned}$$

Evaluating at $t = 0$:

$$\left. \frac{dI}{dt} \right|_{t=0} = I_0 \beta \left(S_0 - \frac{\alpha}{\beta} \right) \quad (\text{iv})$$

From $\frac{dS}{dt} = -\beta SI$, since β, S, I are positive, we have $\frac{dS}{dt} < 0$, which implies $S(t)$ is a strictly decreasing function. Therefore:

$$S(t) < S(0) = S_0 \quad \text{for all } t > 0$$

Case 1: No Epidemic Outbreak $\left(S_0 \leq \frac{\alpha}{\beta} \right)$

From equation (iv):

$$\frac{dI}{dt} = I\beta \left(S - \frac{\alpha}{\beta} \right) < I\beta \left(S_0 - \frac{\alpha}{\beta} \right) \quad \text{since } S(t) < S_0$$

Now, if $S_0 \leq \frac{\alpha}{\beta}$:

$$\begin{aligned}\frac{dI}{dt} &< \beta I \left(\frac{\alpha}{\beta} - \frac{\alpha}{\beta} \right) = 0 \\ \implies \frac{dI}{dt} &< 0 \quad \text{for all } t \geq 0\end{aligned}$$

Thus, $I(t)$ is a strictly decreasing function. Integrating gives:

$$\begin{aligned}[I]_0^t &< 0 \implies I(t) < I_0 \\ \implies \lim_{t \rightarrow \infty} I(t) &< I_0\end{aligned}$$

Hence, no epidemic occurs, and the infection dies out.

Case 2: Epidemic Outbreak $\left(S_0 > \frac{\alpha}{\beta} \right)$

Let $S_0 > \frac{\alpha}{\beta}$. From equation (iv):

$$\begin{aligned}\left. \frac{dI}{dt} \right|_{t=0} &= I_0 \beta \left(S_0 - \frac{\alpha}{\beta} \right) > 0 \\ \implies \left. \frac{dI}{dt} \right|_{t=0} &> 0\end{aligned}$$

Since the derivative is positive at $t = 0$, $I(t)$ is initially an increasing function. Thus, for early times $t > 0$:

$$I(t) > I(0) = I_0$$

The number of infected individuals increases initially. Hence, an **epidemic occurs** and the infection spreads through the population.

7.2 Discuss the deterministic susceptible infected(SI) model and solve. Show that the infection will spread throughout the population.

Solution:

The **SI model** is the simplest mathematical framework for describing the spread of an infectious disease in a closed population without recovery or immunity.

Assumptions:

1. The total population size N is fixed and constant over time:

$$S(t) + I(t) = N$$

where:

- $S(t)$ is the number of **susceptible** individuals at time t .
 - $I(t)$ is the number of **infected** individuals at time t .
2. The rate of new infections is proportional to the product of susceptible individuals $S(t)$ and infected individuals $I(t)$ (mass-action principle).
 3. Once an individual becomes infected, they remain infected indefinitely (no recovery or removal).

Governing Differential Equations:

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI$$

where $\beta > 0$ is the transmission or infection rate parameter. The total population size is given by:

Solving for $S(t)$

Let

$$\text{total population} = S(t) + I(t) = n + 1 \tag{1}$$

From Equation (2), we have:

$$\frac{dS}{dt} = -\beta SI$$

Substituting $I(t) = (n + 1) - S(t)$ from Equation (1):

$$I(t) = (n + 1) - S(t)$$

We get:

$$\begin{aligned}
\frac{dS}{dt} &= -\beta S((n+1) - S) \\
\implies \frac{dS}{-S((n+1) - S)} &= \beta dt \\
\implies -\frac{1}{n+1} \left(\frac{1}{S} + \frac{1}{n+1-S} \right) dS &= \beta dt
\end{aligned}$$

Integrating both sides:

$$\begin{aligned}
-\{\ln S - \ln(n+1-S)\} &= (n+1)\beta t + \ln c \\
\implies \ln \left(\frac{n+1-S}{S} \right) &= (n+1)\beta t + \ln c
\end{aligned}$$

Applying initial conditions ($t=0, S=n$):

$$\ln \left(\frac{1}{n} \right) = \ln c \implies c = \frac{1}{n}$$

Substituting c back into the integral equation:

$$\begin{aligned}
\ln \left(\frac{n+1-S}{S} \right) &= (n+1)\beta t + \ln \left(\frac{1}{n} \right) \\
\implies \ln \left(\frac{\frac{n+1-S}{S}}{\frac{1}{n}} \right) &= (n+1)\beta t \\
\implies \ln \left\{ \frac{n(n+1-S)}{S} \right\} &= (n+1)\beta t \\
\implies \frac{n(n+1-S)}{S} &= e^{(n+1)\beta t} \\
\implies n^2 + n - Sn &= Se^{(n+1)\beta t} \\
\implies S[n + e^{(n+1)\beta t}] &= n(n+1) \\
\implies S(t) &= \frac{n(n+1)}{n + e^{(n+1)\beta t}} \tag{5}
\end{aligned}$$

Solving for $I(t)$

Now, considering the differential equation for infected individuals:

$$\begin{aligned}
\frac{dI}{dt} &= \beta SI = \beta((n+1) - I)I \\
\implies \frac{dI}{I(n+1-I)} &= \beta dt \\
\implies \frac{1}{n+1} \left(\frac{1}{I} + \frac{1}{n+1-I} \right) dI &= \beta dt \\
\implies \ln I - \ln(n+1-I) &= (n+1)\beta t + \ln c \\
\implies \ln \left\{ \frac{I}{n+1-I} \right\} &= (n+1)\beta t + \ln c
\end{aligned}$$

At $t = 0, I = 1$:

$$\ln \left\{ \frac{1}{n+1-1} \right\} = \ln c \implies c = \frac{1}{n}$$

Substituting c :

$$\begin{aligned} \implies \frac{I}{n+1-I} &= \frac{1}{n} e^{(n+1)\beta t} \\ \implies I(t) &= \frac{n+1}{1 + n e^{-(n+1)\beta t}} \end{aligned} \tag{6}$$

3. Limiting Behavior as $t \rightarrow \infty$

Taking limits as $t \rightarrow \infty$:

$$\lim_{t \rightarrow \infty} S(t) = \lim_{t \rightarrow \infty} \frac{n(n+1)}{n + e^{(n+1)\beta t}} = 0$$

From (6):

$$\lim_{t \rightarrow \infty} I(t) = \lim_{t \rightarrow \infty} \frac{n+1}{1 + n e^{-(n+1)\beta t}} = n+1$$

So, the infection will spread throughout the entire population.

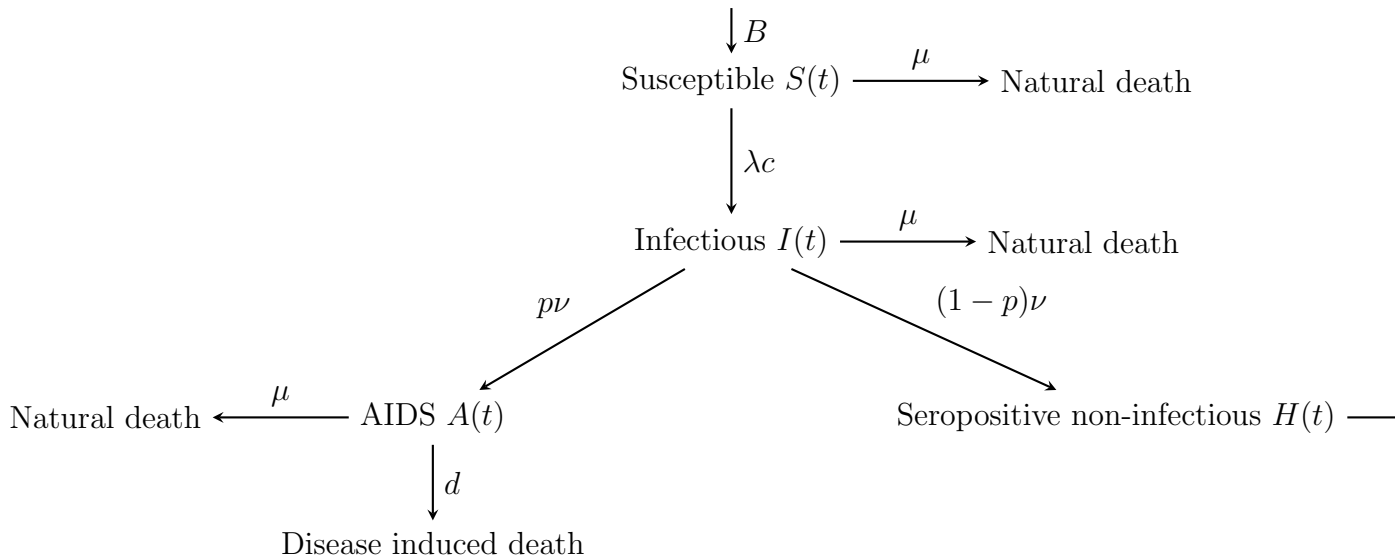
Chapter 8

HIV

8.1 Describe the deterministic AIDS model. Show that the infection spread throughout the population. Find the doubling time. Also find the number of AIDS patients at any time t .

Solution. Let $N(t)$ denote the size of the total population. Let $S(t)$, $I(t)$, $A(t)$ and $H(t)$ denote, respectively, the number of susceptibles, infected population, AIDS patients, and the number of HIV-positive population who are non-infectious.

Now we consider the following flow chart:



Here B is the recruitment of susceptibles, μ is the natural (non-AIDS) death rate, λc is the rate of transferral from susceptible to the infectious class, λ is the probability of acquiring infection from a randomly chosen partner

$$\left(\lambda = \frac{\beta I}{N} \right),$$

where β is the transmission probability, c is the number of partners, d is the AIDS-related death rate, p is the proportion of HIV-positive who are infectious, and ν is the rate of conversion from infection to AIDS, here taken to be constant. $\frac{1}{\nu}$, say $= D$, is the average incubation time of the disease.

Therefore, a reasonable model based on the flow chart is

$$\frac{dS}{dt} = B - \mu S - \lambda c S, \quad \lambda = \frac{\beta I}{N} \quad (1)$$

$$\frac{dI}{dt} = \lambda c S - (\nu + \mu) I \quad (2)$$

$$\frac{dA}{dt} = p\nu I - (d + \mu) A \quad (3)$$

$$\frac{dH}{dt} = (1 - p)\nu I - \mu H \quad (4)$$

$$N(t) = S(t) + I(t) + H(t) + A(t) \quad (5)$$

Now, differentiating (5) w.r.t. t , we get

$$\begin{aligned} \frac{dN}{dt} &= \frac{dS}{dt} + \frac{dI}{dt} + \frac{dH}{dt} + \frac{dA}{dt} \\ &= [B - \mu S - \lambda c S] + [\lambda c S - (\nu + \mu) I] + [(1 - p)\nu I - \mu H] + [p\nu I - (d + \mu) A] \\ &= B - \mu S - \mu I - \mu H - (d + \mu) A \\ &= B - \mu(S + I + H + A) - dA \\ \Rightarrow \frac{dN}{dt} &= B - \mu N - dA. \end{aligned} \quad (6)$$

Now, if at $t = 0$ an infected individual is introduced into an otherwise infection-free population of susceptibles, then we have, near $t = 0$, $N \approx S$. Therefore, from (2), we get

$$\begin{aligned} \frac{dI}{dt} &\approx \lambda c N - (\nu + \mu) I, \quad \text{since } N \approx S \\ \frac{dI}{dt} &\approx c \left(\frac{\beta I}{N} \right) N - (\nu + \mu) I \\ \Rightarrow \frac{dI}{dt} &\approx (\beta c - \nu - \mu) I \\ \Rightarrow \frac{dI}{dt} &\approx \nu \left(\frac{\beta c}{\nu} - \frac{\mu}{\nu} - 1 \right) I \end{aligned} \quad (7)$$

Since the average incubation time $\frac{1}{\nu}$ is very much shorter than the average life expectancy $\frac{1}{\mu}$, so $\frac{\mu}{\nu} \approx 0$. Therefore, from (7), we get

$$\frac{dI}{dt} \approx \nu(R_0 - 1)I, \quad \text{where } R_0 \approx \frac{\beta c}{\nu} > 1.$$

Thus the basic reproductive rate R_0 is given in terms of the number of partners c , the transmission probability β , and the average incubation time of the disease $\frac{1}{\nu}$.

Now, we can get some interesting information from the analysis of the system (1)–(5) during the early stages of an epidemic. Here the population consists of almost all

susceptibles, and so $N \approx S$. Therefore, the growth rate of infection, i.e. the HIV-positive I class, is approximated by equation (8). The solution of (8) is

$$I(t) = I(0) e^{\nu(R_0-1)t} = I(0) e^{rt}, \quad (10)$$

where $r = \nu(R_0 - 1)$ is positive only if an epidemic exists (i.e. $R_0 > 1$), and $I(0)$ is the initial number of people introduced into the susceptible population.

From (10), we observe that $I(t)$, i.e. infection, grows exponentially, and ultimately the whole population will be infected.

Doubling time. Suppose the doubling time is t_d , i.e. $I(t_d) = 2I(0)$. From (10), we get

$$t_d = \frac{1}{r} \ln 2 = \frac{\ln 2}{\nu(R_0 - 1)}. \quad (11)$$

Thus, we observe that the larger the basic reproductive rate R_0 , the shorter the doubling time.

Number of AIDS patients at any time t . Substituting (10) into equation (3), we get

$$\frac{dA}{dt} = p\nu I_0 e^{rt} - (d + \mu)A. \quad (12)$$

At the initial stage of the epidemic there were no AIDS patients, so $A(0) = 0$. Therefore, the solution of (12) is

$$A(t) = p\nu I_0 \frac{e^{rt} - e^{-(d+\mu)t}}{r + d + \mu}, \quad (13)$$

which is the number of AIDS patients at any time t .

8.2 For understanding

$$\begin{aligned} \frac{dA}{dt} &= p\nu I_0 e^{rt} - (d + \mu)A. \\ \frac{dA}{dt} + (d + \mu)A &= p\nu I_0 e^{rt}. \\ \text{I.F.} &= e^{\int (d+\mu) dt} = e^{(d+\mu)t}. \end{aligned} \quad (12)$$

Multiplying both sides by the integrating factor:

$$\begin{aligned} \frac{d}{dt} [A(t)e^{(d+\mu)t}] &= p\nu I_0 e^{[r+(d+\mu)]t}. \\ A(t)e^{(d+\mu)t} &= \frac{p\nu I_0}{r + d + \mu} e^{[r+(d+\mu)]t} + C, \\ A(t) &= \frac{p\nu I_0}{r + d + \mu} e^{rt} + Ce^{-(d+\mu)t}. \end{aligned}$$

Using the initial condition at the start of the epidemic, there were no AIDS patients ($A(0) = 0$):

$$\begin{aligned} 0 &= \frac{p\nu I_0}{r + d + \mu} + C \implies C = -\frac{p\nu I_0}{r + d + \mu}. \\ A(t) &= p\nu I_0 \left[\frac{e^{rt} - e^{-(d+\mu)t}}{r + d + \mu} \right]. \end{aligned} \quad (13)$$